

TEXAS A&M UNIVERSITY

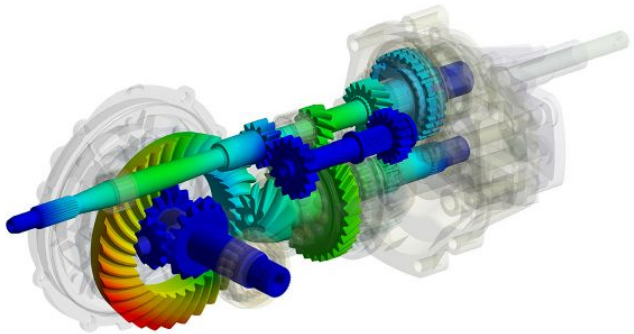
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

FEA II

Mechanical Incubator

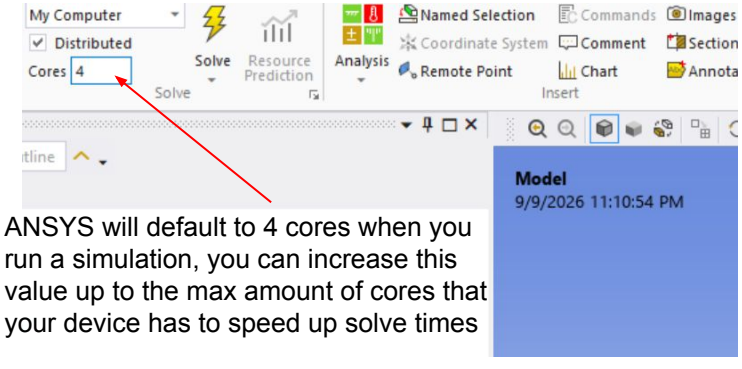
Today's Agenda:

- Intro
- In class simulations and instruction slides:
-
- Fatigue Analysis On mesh plate with hole
- Antweight chassis structural analysis
- Explicit Dynamic Simulation car crash
-
- Topics to cover:
- What each analysis is in ansys
- Examples of each analysis
- How it can be applied real world
- Optimization of ansys

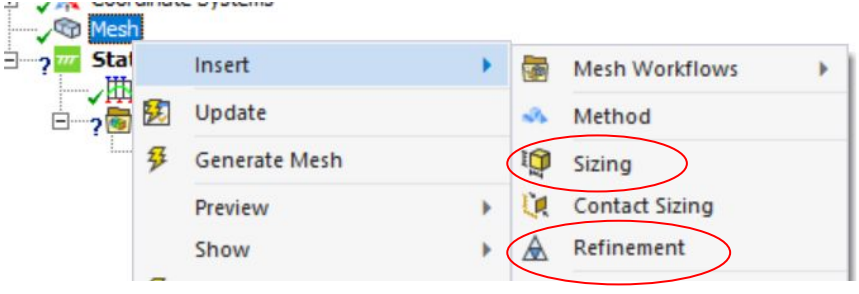


ANSYS Optimizations

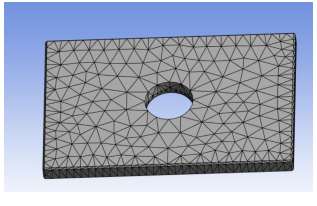
Here are a couple ways to improve the simulation time of your ANSYS Simulations



ANSYS will default to 4 cores when you run a simulation, you can increase this value up to the max amount of cores that your device has to speed up solve times



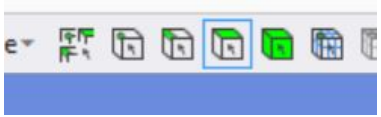
Refinement and Sizing Meshes can both increase the quality of the mesh in specified areas. Refinement will subdivide the mesh a certain amount of times while sizing allows you to pick a specific element size

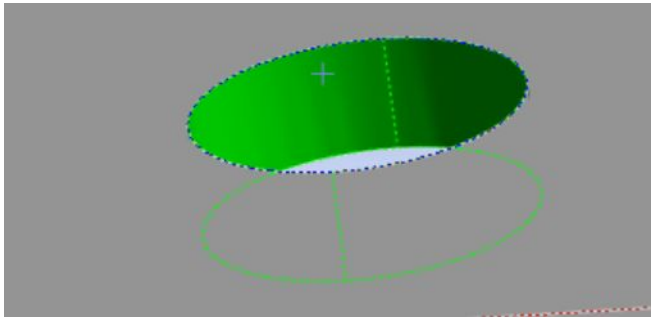


<- Before

ANSYS Optimizations

Use the selection tool to select the desired geometry (faces in this case but can be edges or corners as well)

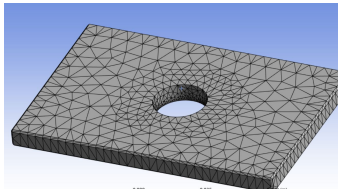




Details of "Refinement" - Refinement

Scope	
Scoping Method	Geometry Selection
Geometry	2 Faces
Definition	
Suppressed	No
Refinement	1

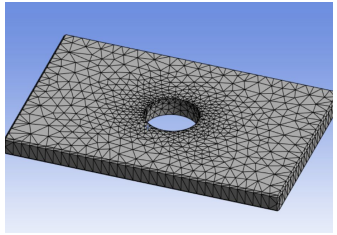
The refinement number will determine how many times each element will be subdivided



Details of "Face Sizing" - Sizing

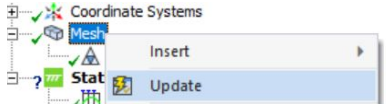
Scope	
Scoping Method	Geometry Selection
Geometry	2 Faces
Definition	
Suppressed	No
Type	Element Size
Element Size	Default (8.9995e-003 m)
Advanced	
Defeature Size	Default
Influence Volume	No
Behavior	Soft

Element size will determine how large each element is.



Element size of 4e-3m

Make sure to update mesh after done





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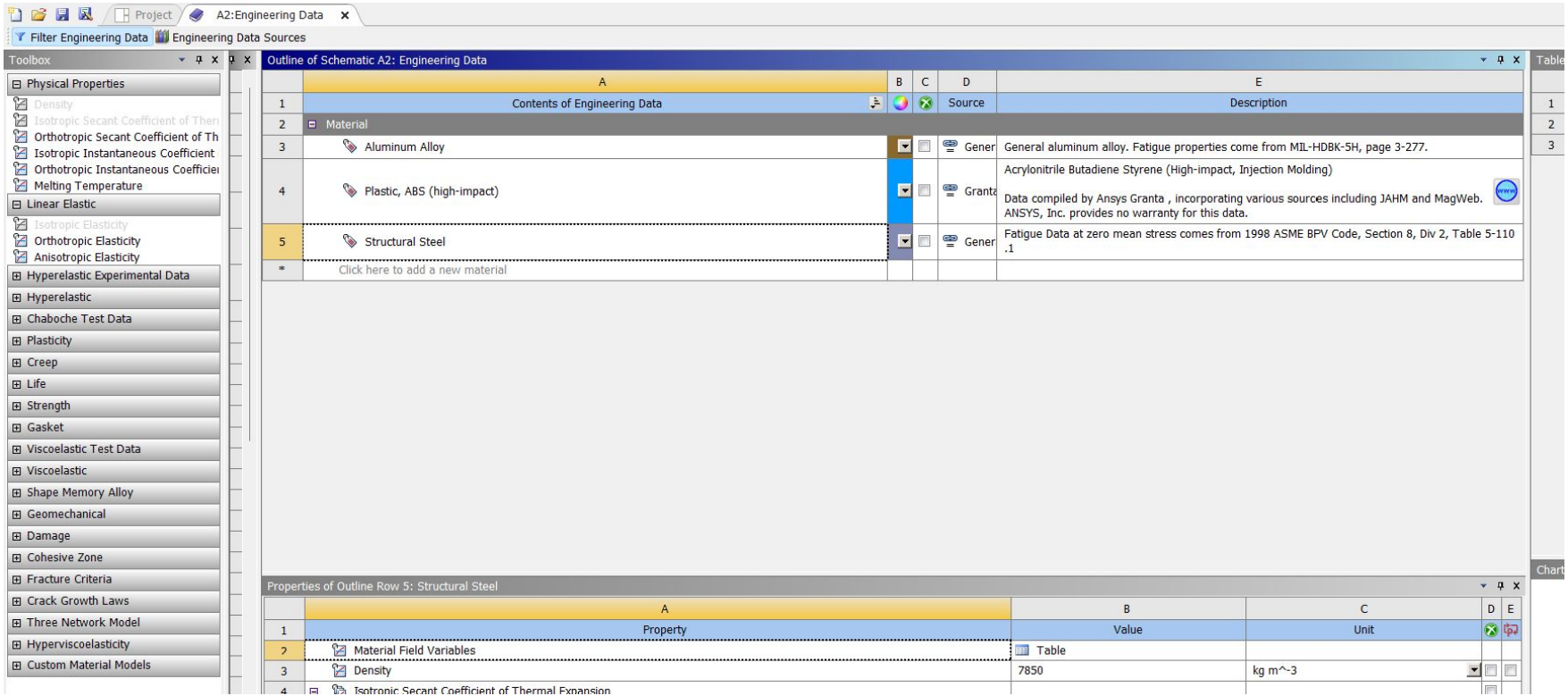
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

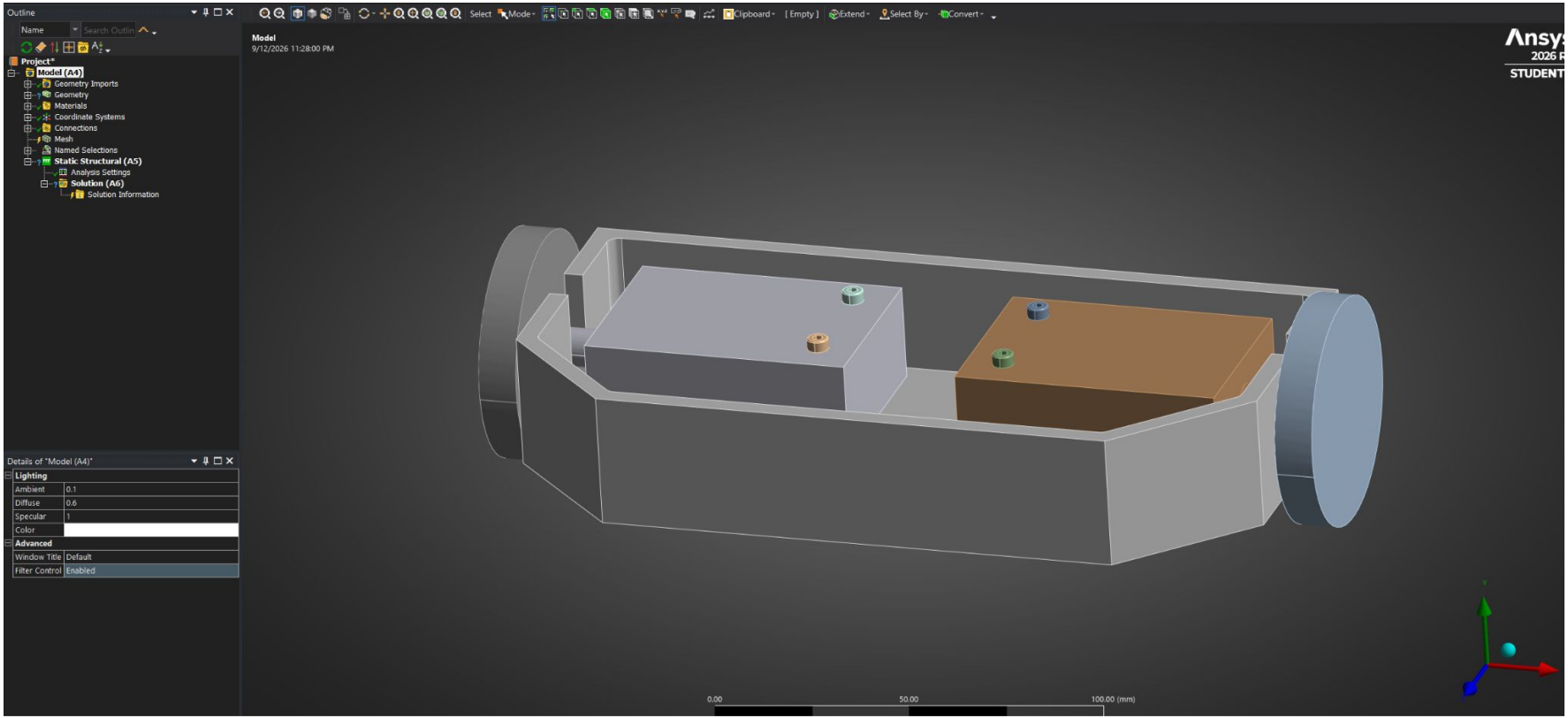
Structural Analysis on Antweight Chassis

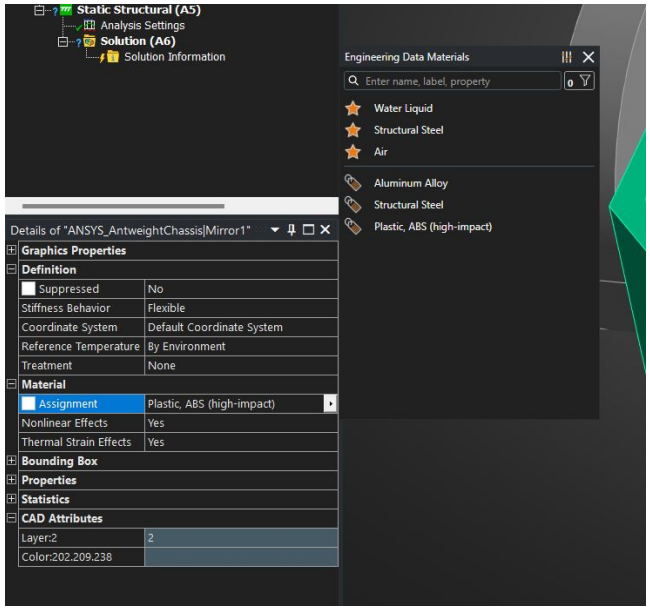
Why Static Structural Analysis on an Assembly?

- Last week we learned how to do a static structural analysis on a singular part.

  However there will often be times where it is difficult to predict where forces will be present in a complex assembly, so it would be better to simulate the whole assembly instead of individual pieces.

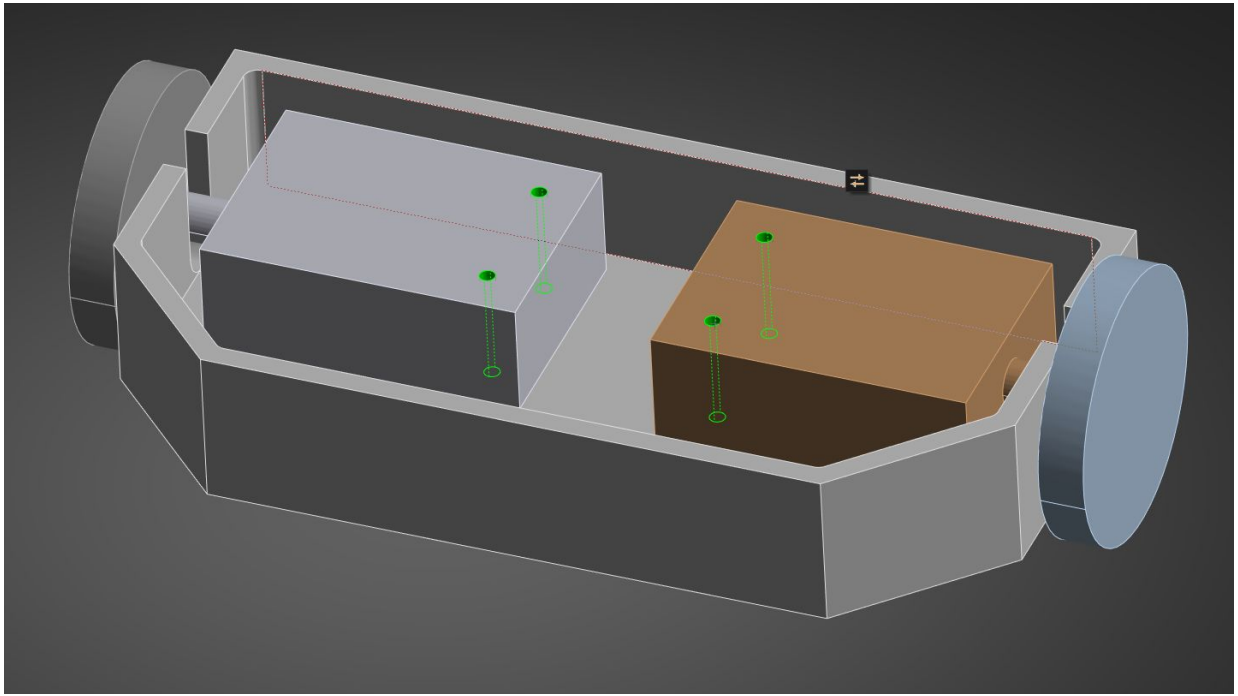

Project
A2:Engineering Data
Filter Engineering Data
Engineering Data Sources
Toolbox
Physical Properties
Density
Isotropic Secant Coefficient of Therl
Orthotropic Secant Coefficient of Th
Isotropic Instantaneous Coefficient
Orthotropic Instantaneous Coefficie
Melting Temperature
Linear Elastic
Isotropic Elasticity
Orthotropic Elasticity
Anisotropic Elasticity
Hyperelastic Experimental Data
Hyperelastic
Chaboche Test Data
Plasticity
Creep
Life
Strength
Gasket
Viscoelastic Test Data
Viscoelastic
Shape Memory Alloy
Geomechanical
Damage
Cohesive Zone
Fracture Criteria
Crack Growth Laws
Three Network Model
Hyperviscoelasticity
Custom Material Models
Outline of Schematic A2: Engineering Data
Contents of Engineering Data
Source
Description
Material
Aluminum Alloy
General aluminum alloy. Fatigue properties come from MIL-HDBK-5H, page 3-277.
Plastic, ABS (high-impact)
Acrylonitrile Butadiene Styrene (High-impact, Injection Molding)
Data compiled by Ansys Granta , incorporating various sources including JAHM and MagWeb. ANSYS, Inc. provides no warranty for this data.
Structural Steel
Fatigue Data at zero mean stress comes from 1998 ASME BPV Code, Section 8, Div 2, Table 5-110 .1
Click here to add a new material
Properties of Outline Row 5: Structural Steel
Property
Value
Unit
Material Field Variables
Table
Density
7850
kg m^-3
Isotropic Secant Coefficient of Thermal Expansion

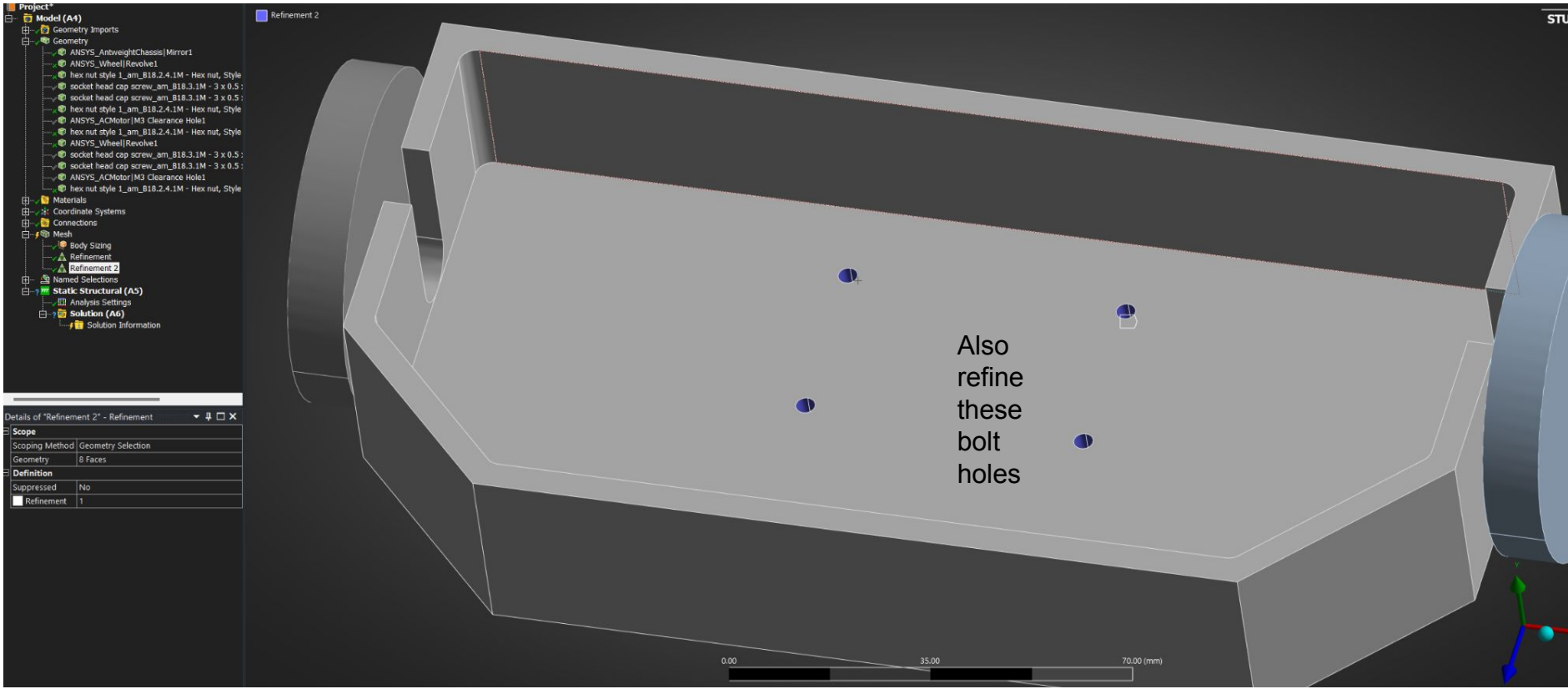

Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
Materials
Coordinate Systems
Connections
Mesh
Named Selections
Static Structural (A5)
Analysis Settings
Solution (A6)
Solution Information
Details of "Model (A4)"
Lighting
Ambient 0.1
Diffuse 0.6
Specular 1
Color
Advanced
Window Title Default
Filter Control Enabled
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
Model
9/12/2026 11:28:00 PM
Ansys
2026 R
STUDENT
0.00
50.00
100.00 (mm)

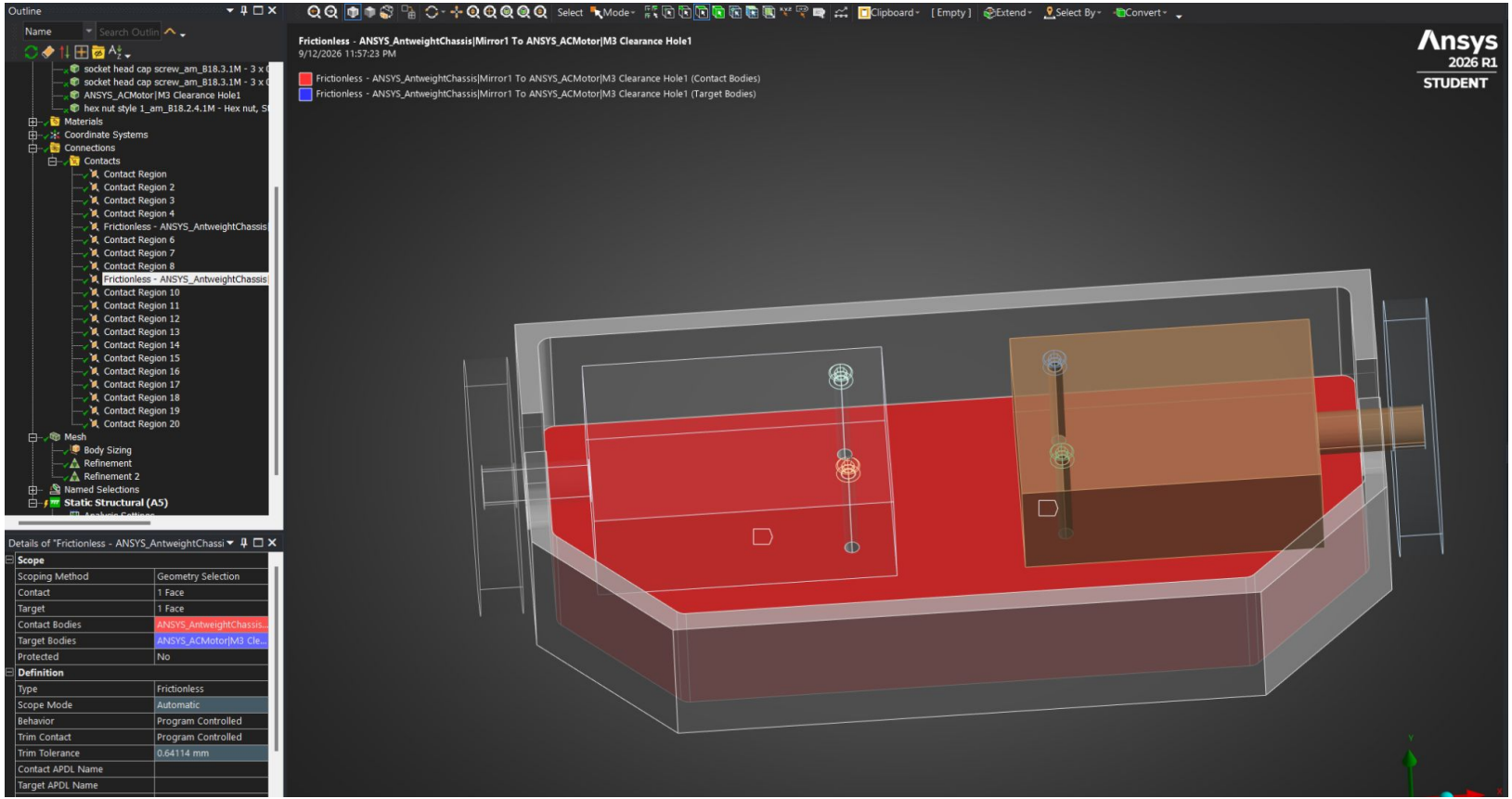


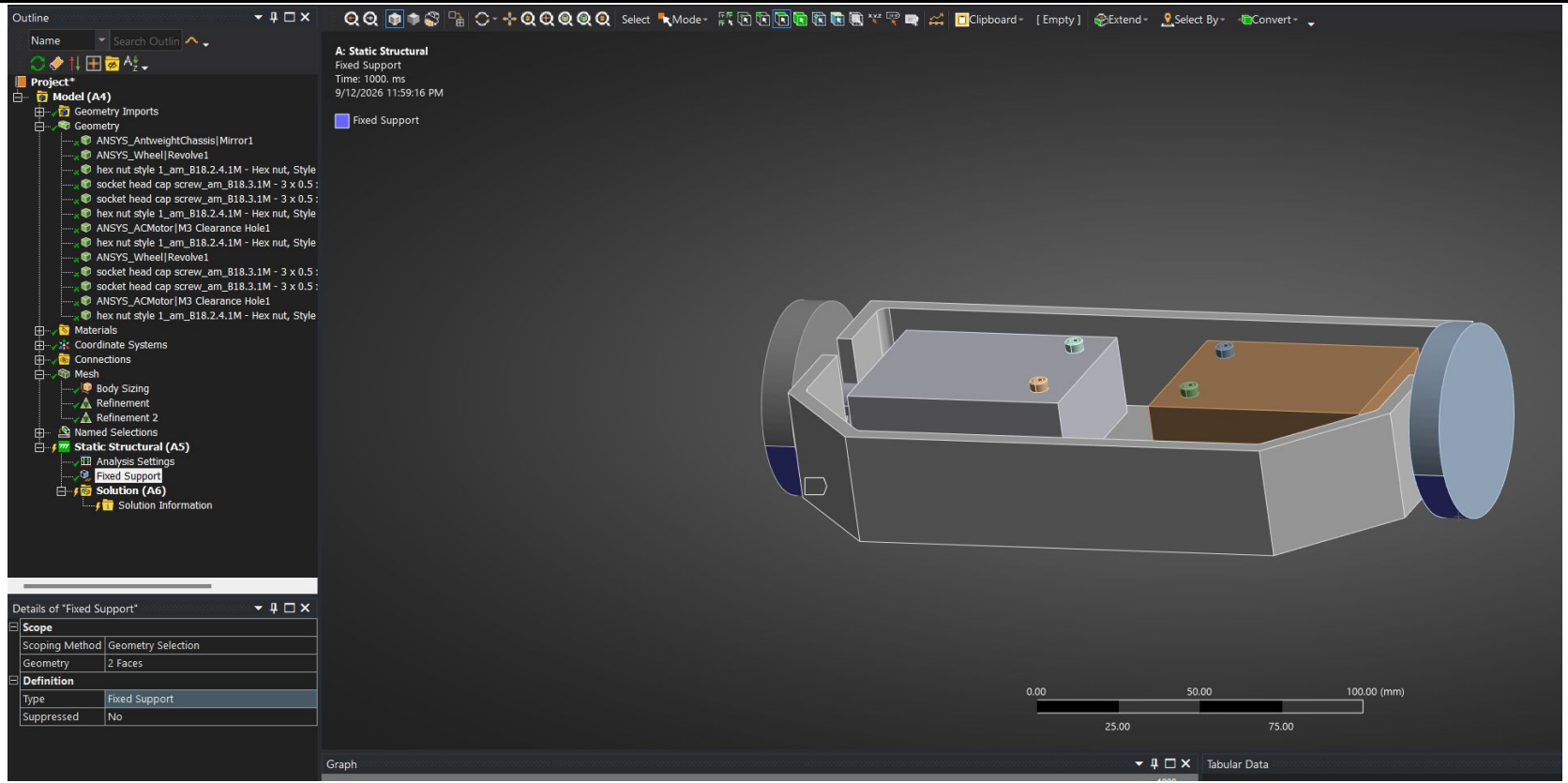
Chassis -> ABS
Wheels -> ABS
Motors -> Aluminum Alloy
Screws/Nuts -> Structural Steel

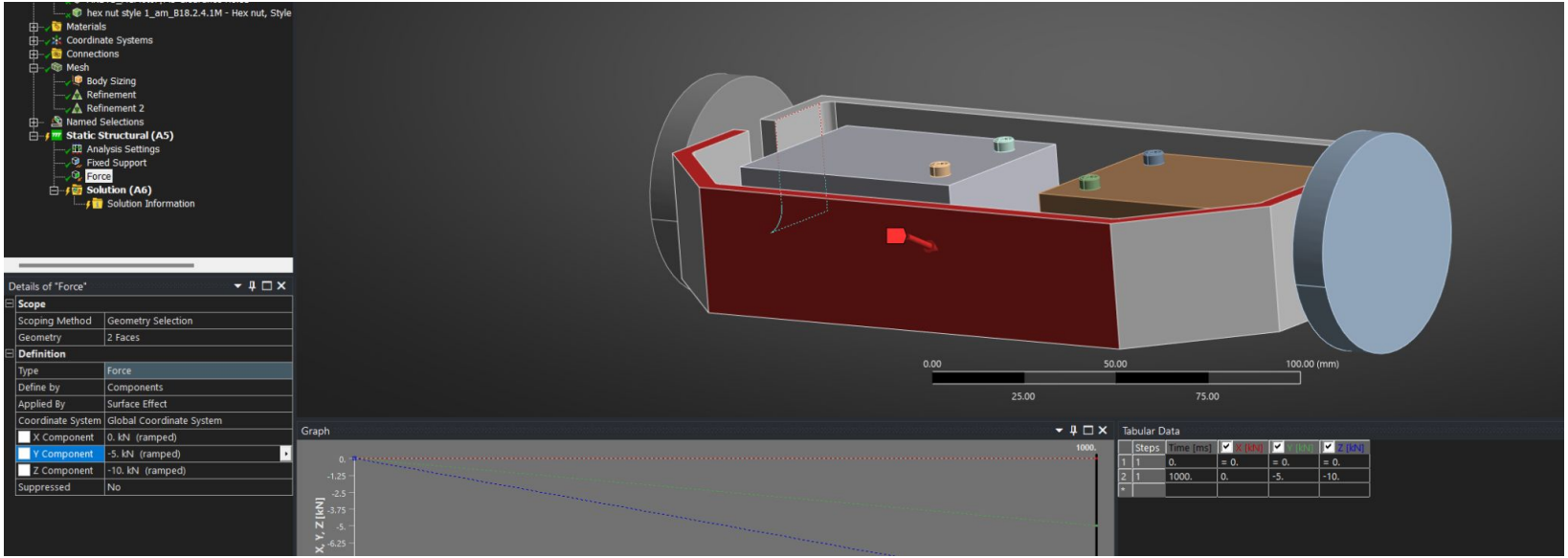
Select the 4 bolt holes and refine the mesh around these areas

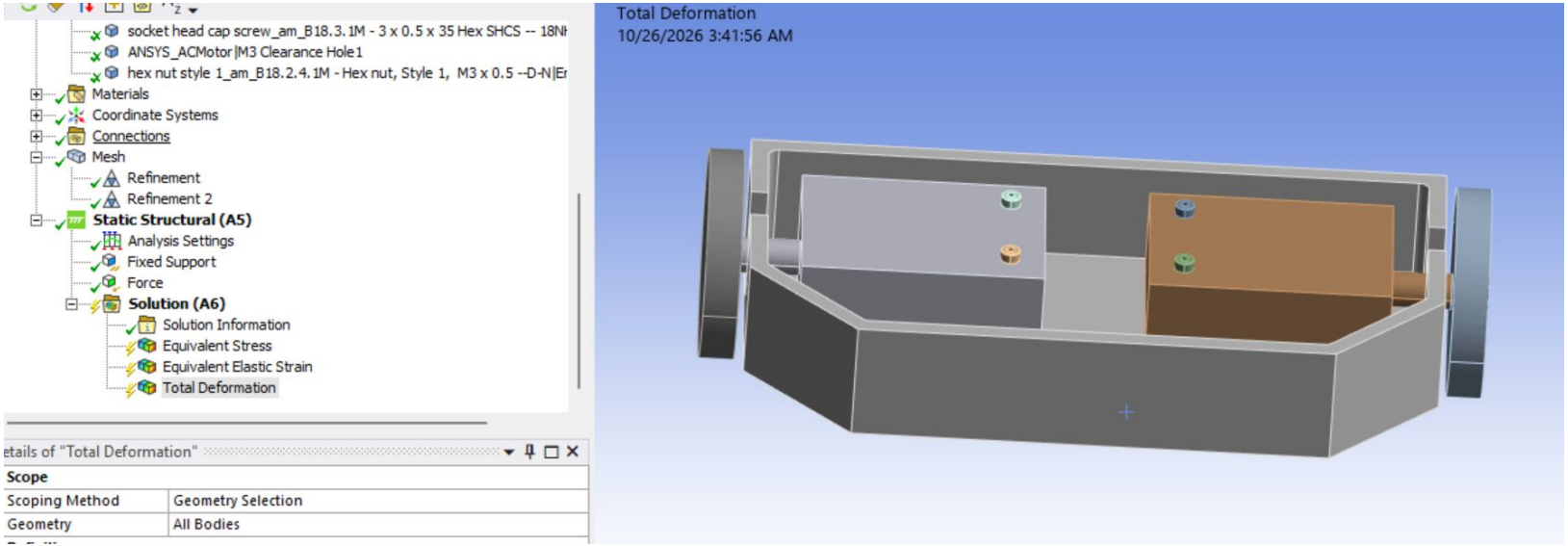



Project*
Model (A4)
Geometry Imports
Geometry
ANSYS_AntweightChassis|Mirror1
ANSYS_Wheel|Revolve1
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
socket head cap screw_am_B18.3.1M - 3 x 0.5 :
socket head cap screw_am_B18.3.1M - 3 x 0.5 :
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
ANSYS_ACMotor|M3 Clearance Hole1
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
ANSYS_Wheel|Revolve1
socket head cap screw_am_B18.3.1M - 3 x 0.5 :
socket head cap screw_am_B18.3.1M - 3 x 0.5 :
ANSYS_ACMotor|M3 Clearance Hole1
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
Materials
Coordinate Systems
Connections
Mesh
Body Sizing
Refinement
Refinement 2
Named Selections
Static Structural (A5)
Analysis Settings
Solution (A6)
Solution Information
Details of "Refinement 2" - Refinement
Scope
Scoping Method
Geometry Selection
Geometry
8 Faces
Definition
Suppressed
No
Refinement
1
Refinement 2
Also refine these bolt holes
0.00
35.00
70.00 (mm)

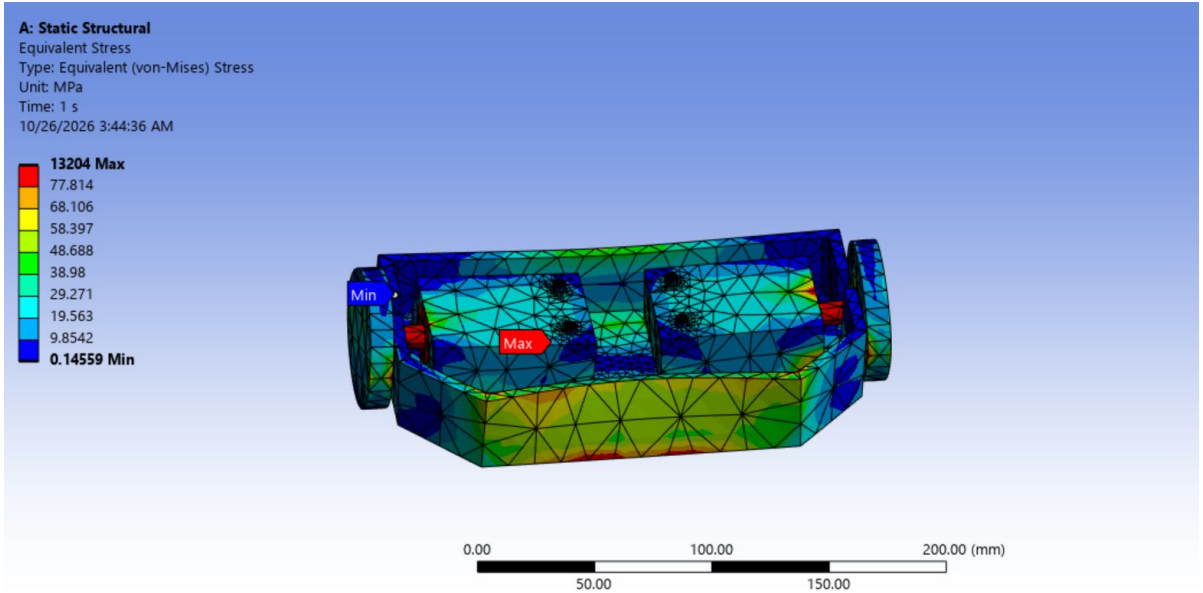

Outline
Name
Search Outlin
socket head cap screw_am_B18.3.1M - 3 x
socket head cap screw_am_B18.3.1M - 3 x
ANSYS_ACMotor|M3 Clearance Hole1
hex nut style 1_am_B18.2.4.1M - Hex nut, S
Materials
Coordinate Systems
Connections
Contacts
Contact Region
Contact Region 2
Contact Region 3
Contact Region 4
Frictionless - ANSYS_AntweightChassis
Contact Region 6
Contact Region 7
Contact Region 8
Frictionless - ANSYS_AntweightChassis
Contact Region 10
Contact Region 11
Contact Region 12
Contact Region 13
Contact Region 14
Contact Region 15
Contact Region 16
Contact Region 17
Contact Region 18
Contact Region 19
Contact Region 20
Mesh
Body Sizing
Refinement
Refinement 2
Named Selections
Static Structural (A5)
Details of "Frictionless - ANSYS_AntweightChassi
Scope
Scoping Method
Geometry Selection
Contact
1 Face
Target
1 Face
Contact Bodies
ANSYS_AntweightChassis...
Target Bodies
ANSYS_ACMotor|M3 Cle...
Protected
No
Definition
Type
Frictionless
Scope Mode
Automatic
Behavior
Program Controlled
Trim Contact
Program Controlled
Trim Tolerance
0.64114 mm
Contact APDL Name
Target APDL Name
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
Frictionless - ANSYS_AntweightChassis|Mirror1 To ANSYS_ACMotor|M3 Clearance Hole1
9/12/2026 11:57:23 PM
Frictionless - ANSYS_AntweightChassis|Mirror1 To ANSYS_ACMotor|M3 Clearance Hole1 (Contact Bodies)
Frictionless - ANSYS_AntweightChassis|Mirror1 To ANSYS_ACMotor|M3 Clearance Hole1 (Target Bodies)
Ansys
2026 R1
STUDENT

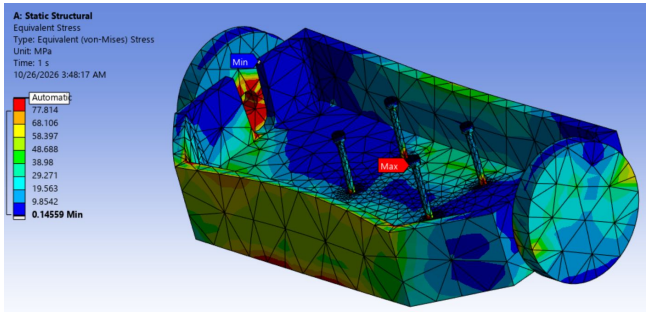

Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
ANSYS_AntweightChassis|Mirror1
ANSYS_Wheel|Revolve1
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
socket head cap screw_am_B18.3.1M - 3 x 0.5
socket head cap screw_am_B18.3.1M - 3 x 0.5
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
ANSYS_ACMotor|M3 Clearance Hole1
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
ANSYS_Wheel|Revolve1
socket head cap screw_am_B18.3.1M - 3 x 0.5
socket head cap screw_am_B18.3.1M - 3 x 0.5
ANSYS_ACMotor|M3 Clearance Hole1
hex nut style 1_am_B18.2.4.1M - Hex nut, Style
Materials
Coordinate Systems
Connections
Mesh
Body Sizing
Refinement
Refinement 2
Named Selections
Static Structural (A5)
Analysis Settings
Fixed Support
Solution (A6)
Solution Information
Details of "Fixed Support"
Scope
Scoping Method
Geometry Selection
Geometry
2 Faces
Definition
Type
Fixed Support
Suppressed
No
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Fixed Support
Time: 1000. ms
9/12/2026 11:59:16 PM
Fixed Support
0.00
50.00
100.00 (mm)
25.00
75.00
Graph
Tabular Data


hex nut style 1_am_B18.2.4.1M - Hex nut, Style
Materials
Coordinate Systems
Connections
Mesh
Body Sizing
Refinement
Refinement 2
Named Selections
Static Structural (A5)
Analysis Settings
Fixed Support
Force
Solution (A6)
Solution Information
Details of "Force"
Scope
Scoping Method
Geometry Selection
Geometry
2 Faces
Definition
Type
Force
Define by
Components
Applied By
Surface Effect
Coordinate System
Global Coordinate System
X Component
0. kN (ramped)
Y Component
-5. kN (ramped)
Z Component
-10. kN (ramped)
Suppressed
No
0.00
50.00
100.00 (mm)
25.00
75.00
Graph
Tabular Data
Steps
Time [ms]
Z [kN]
1
1
0.
0.
0.
0.
2
1
1000.
0.
-5.
-10.
X, Y, Z [kN]

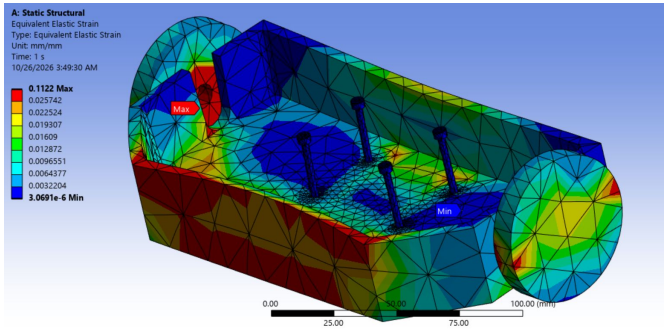


Add solutions, then solve!


A: Static Structural
Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: MPa
Time: 1 s
10/26/2026 3:44:36 AM
13204 Max
77.814
68.106
58.397
48.688
38.98
29.271
19.563
9.8542
0.14559 Min
Min
Max
0.00
100.00
200.00 (mm)
50.00
150.00



If we hide the motors, we can see that there is lots of stress concentrated on the bolts



However, due to them being steel, they have much less strain than the ABS chassis and wheels



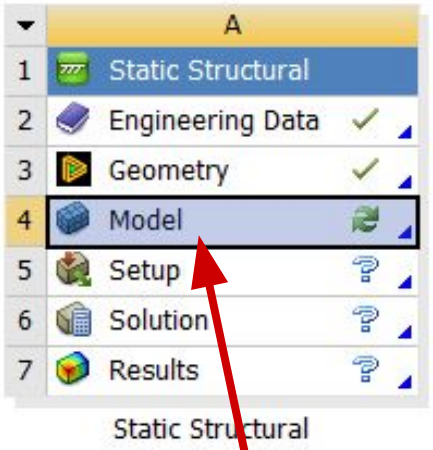
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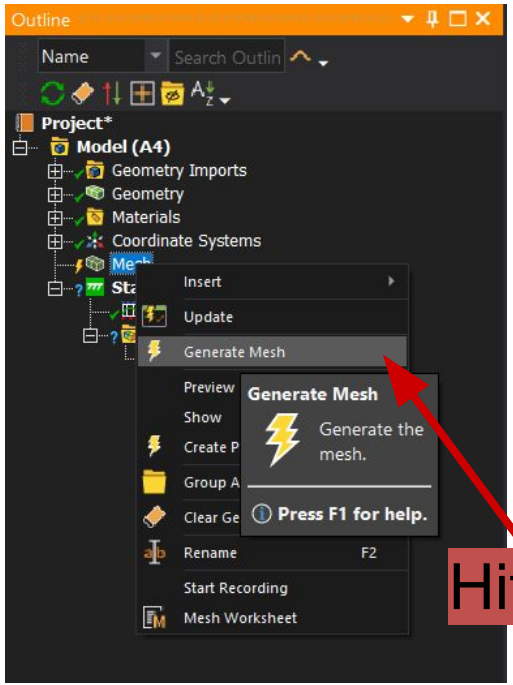
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

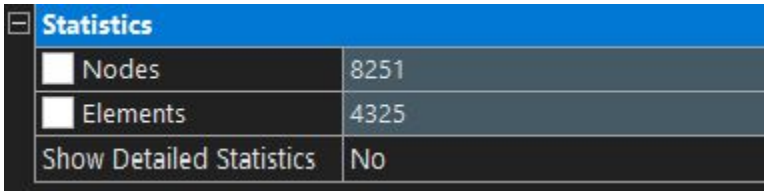
Fatigue Analysis

Why Fatigue Analysis?

- Many applications require repeated cyclic loading over time, rather than a single load occurrence. Doing a fatigue analysis on a part can allow you to simulate how it will behave over this time period, rather than over a single event.





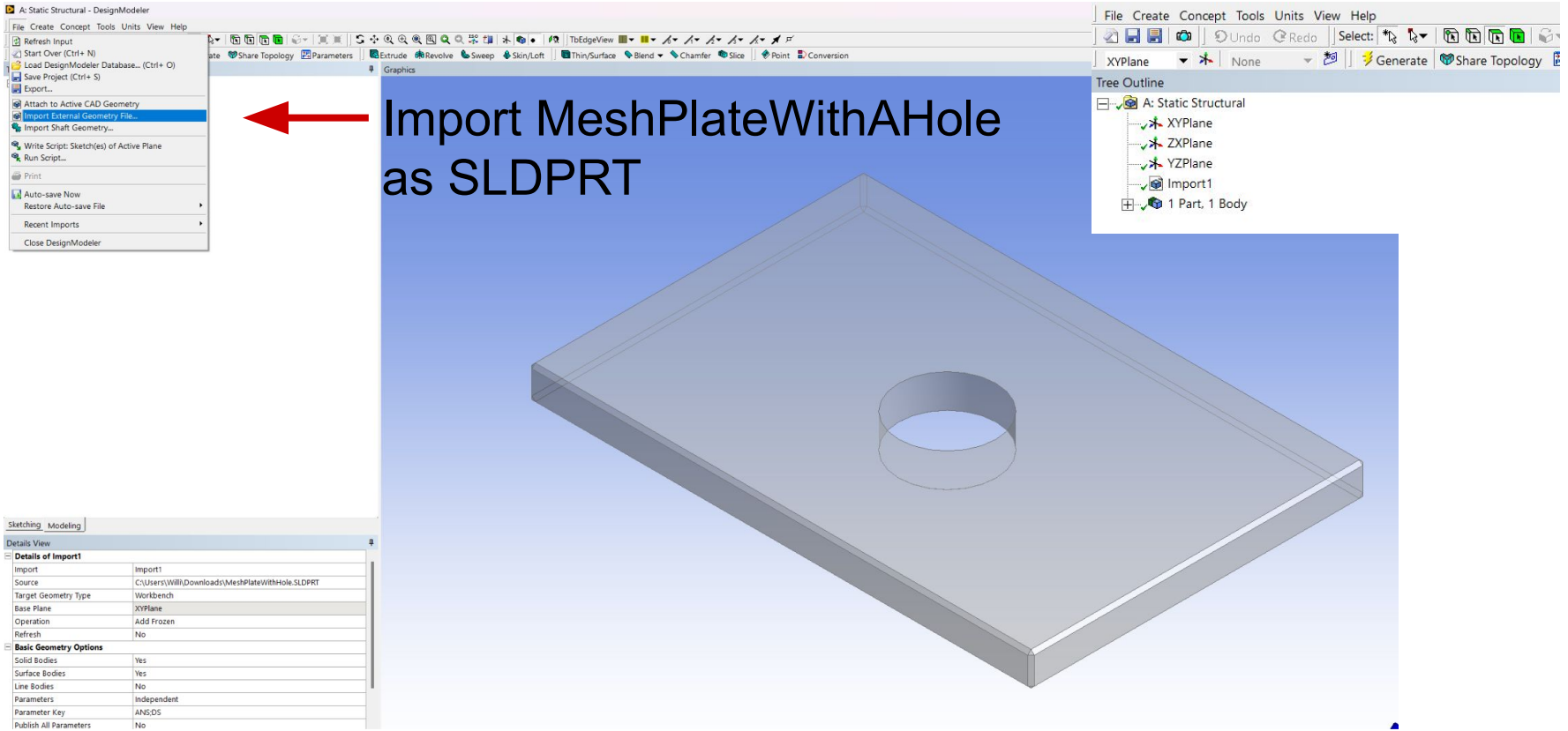


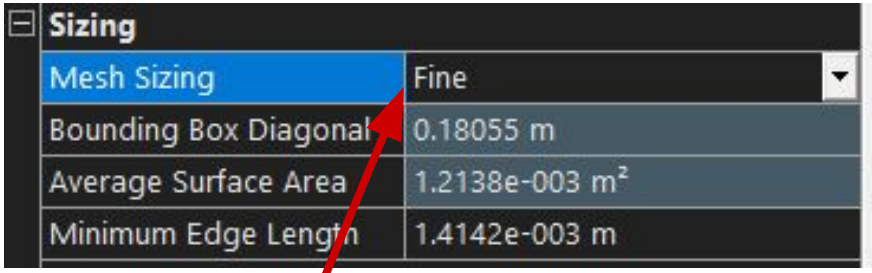
Nodes are individual coordinates where calculations are made, elements connect those nodes in "faces"

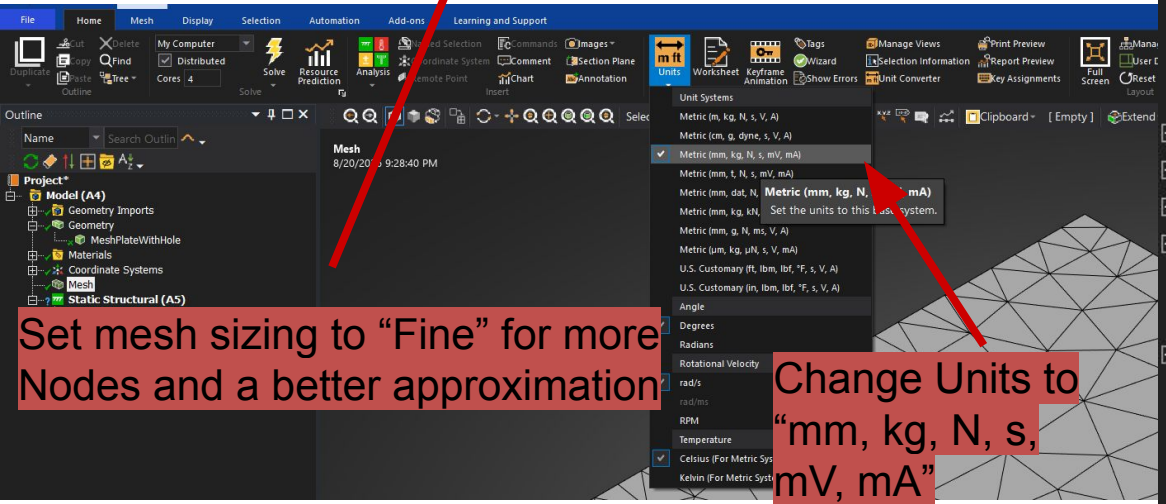
Hit Generate Mesh

Double click model, make sure to open ANSYS Workbench, NOT Discovery

Fatigue Analysis





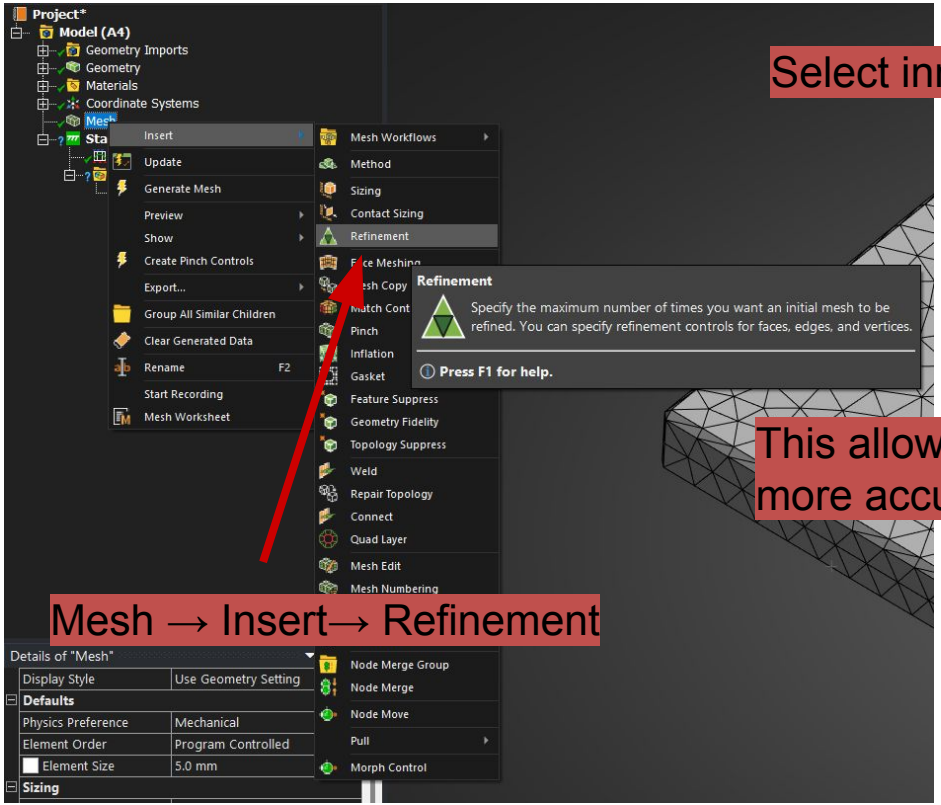


Set mesh sizing to “Fine” for more Nodes and a better approximation

Change Units to “mm, kg, N, s, mV, mA”

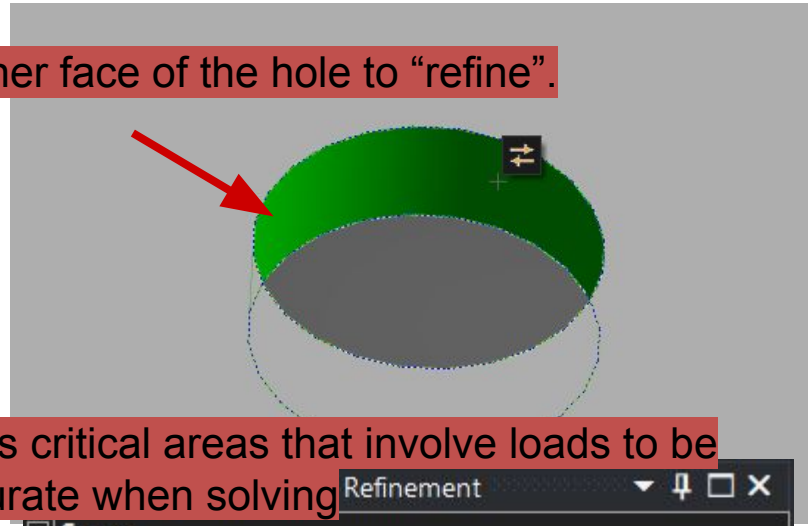
Details of "Mesh"

Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	5.0 mm
Sizing	
Mesh Sizing	Fine
Bounding Box Diagonal	180.55 mm
Average Surface Area	1213.8 mm²
Minimum Edge Length	1.4142 mm
Quality	
Inflation	
Advanced	
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	17343
Elements	10448
Show Detailed Statistics	No



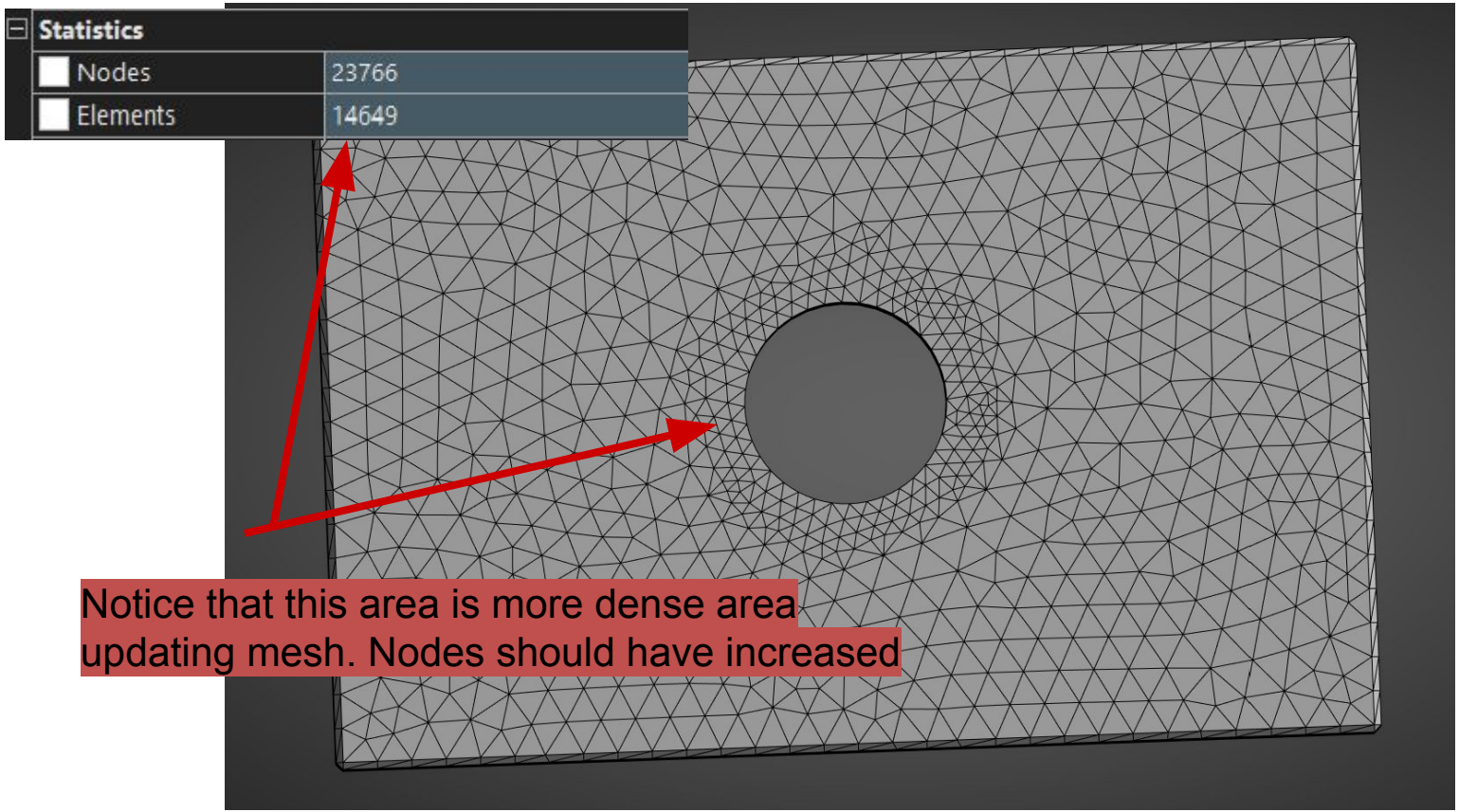
Mesh → Insert→ Refinement

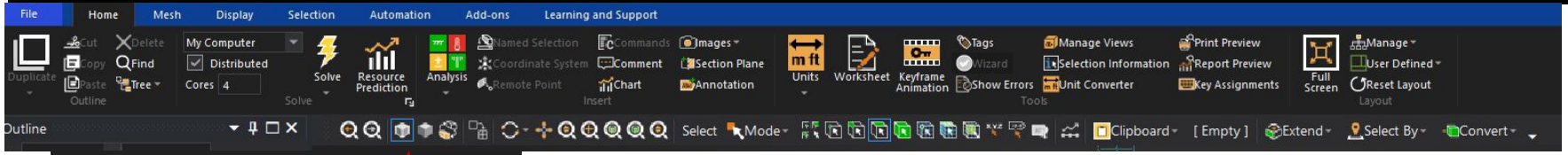
Select inner face of the hole to “refine”.

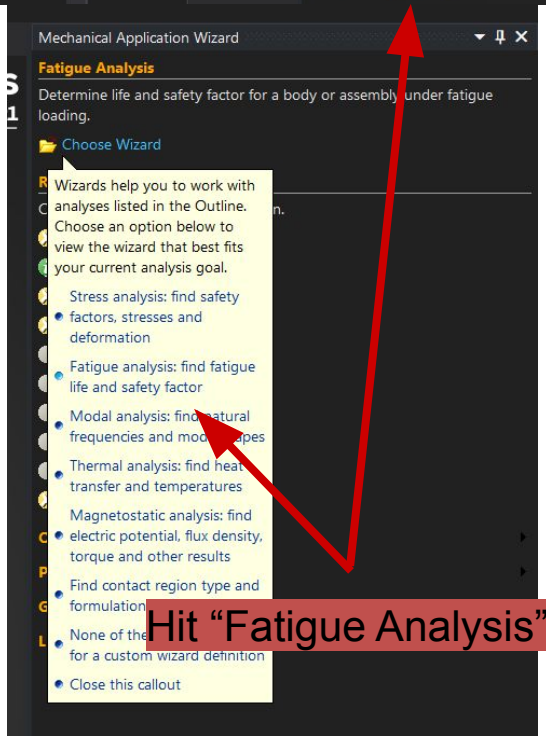


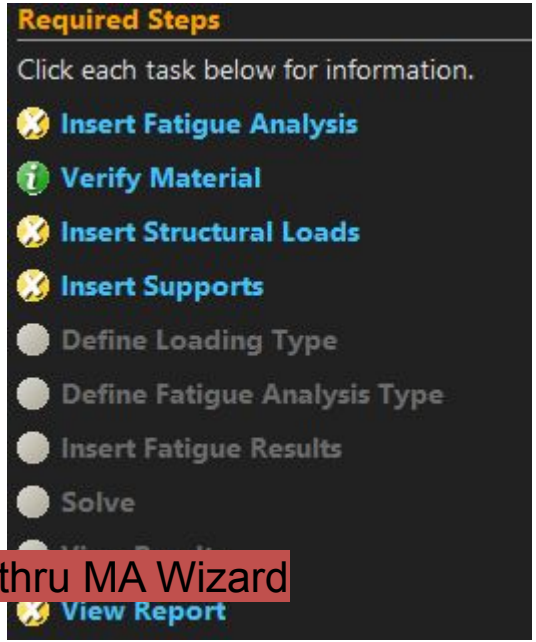
This allows critical areas that involve loads to be more accurate when solving

Refinement	
Scope	
Scoping Method	Geometry Selection
Geometry	1 Face
Definition	
Suppressed	No
Refinement	1


Statistics
Nodes
23766
Elements
14649
Notice that this area is more dense area
updating mesh. Nodes should have increased

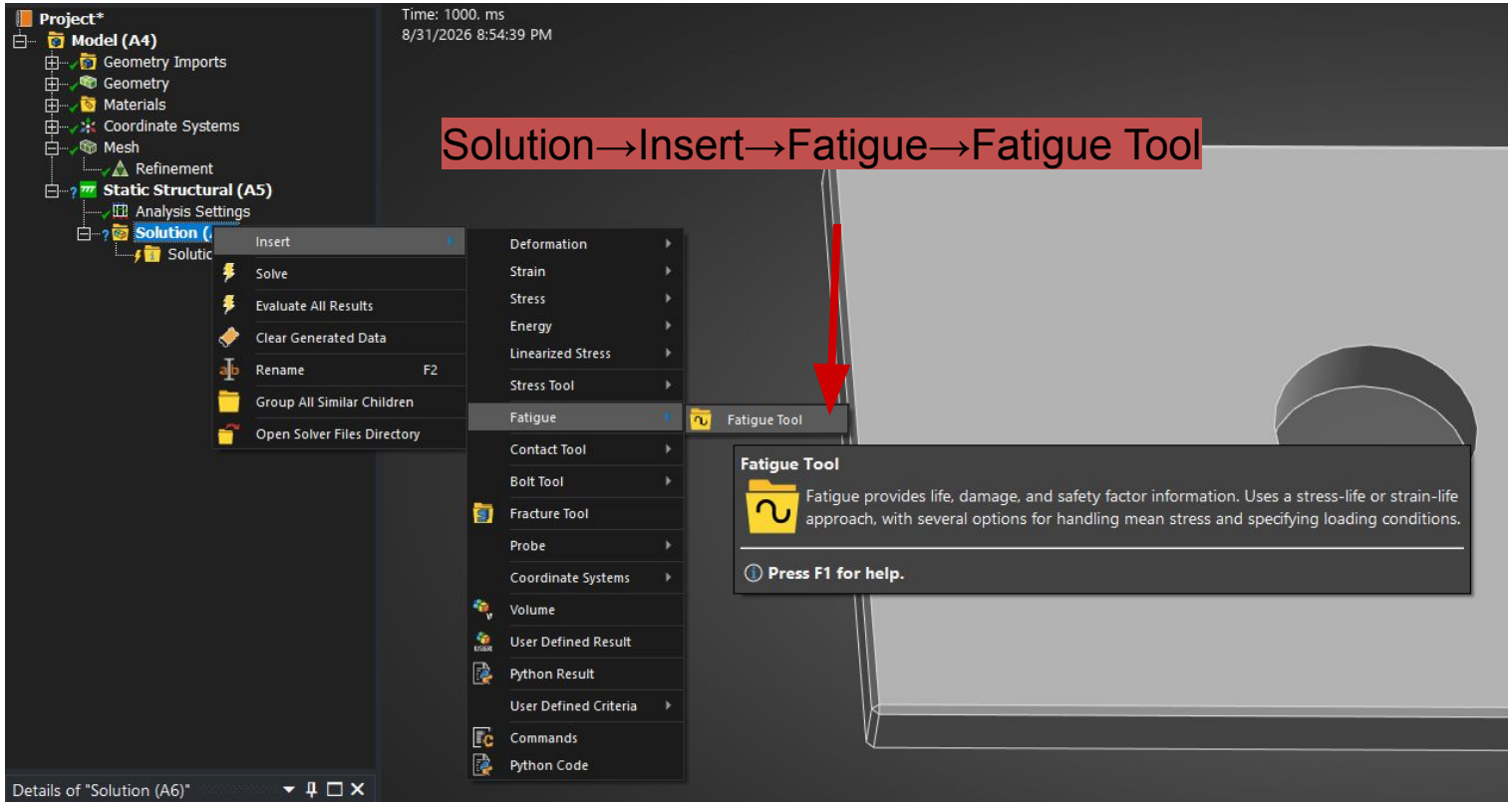


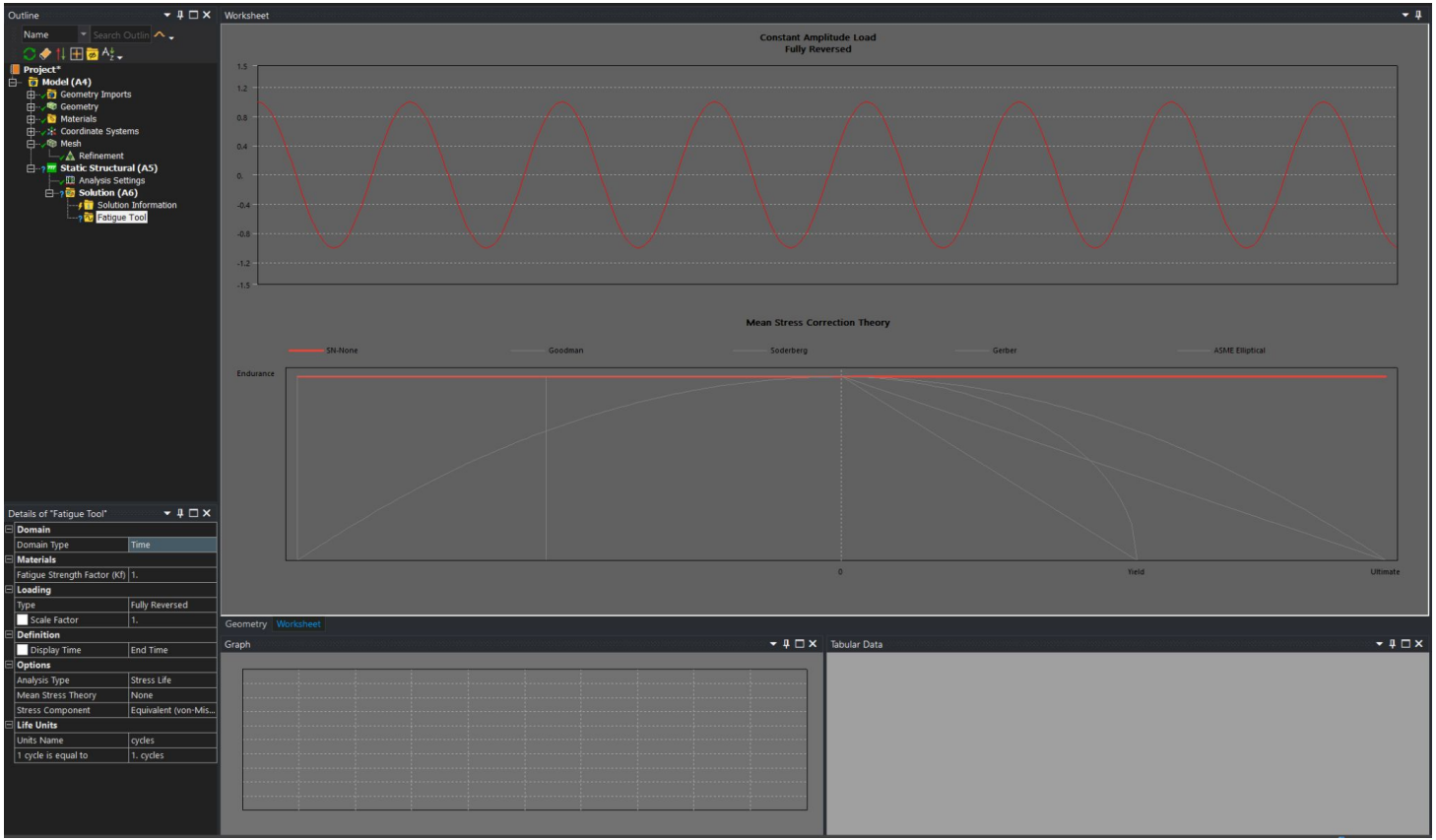


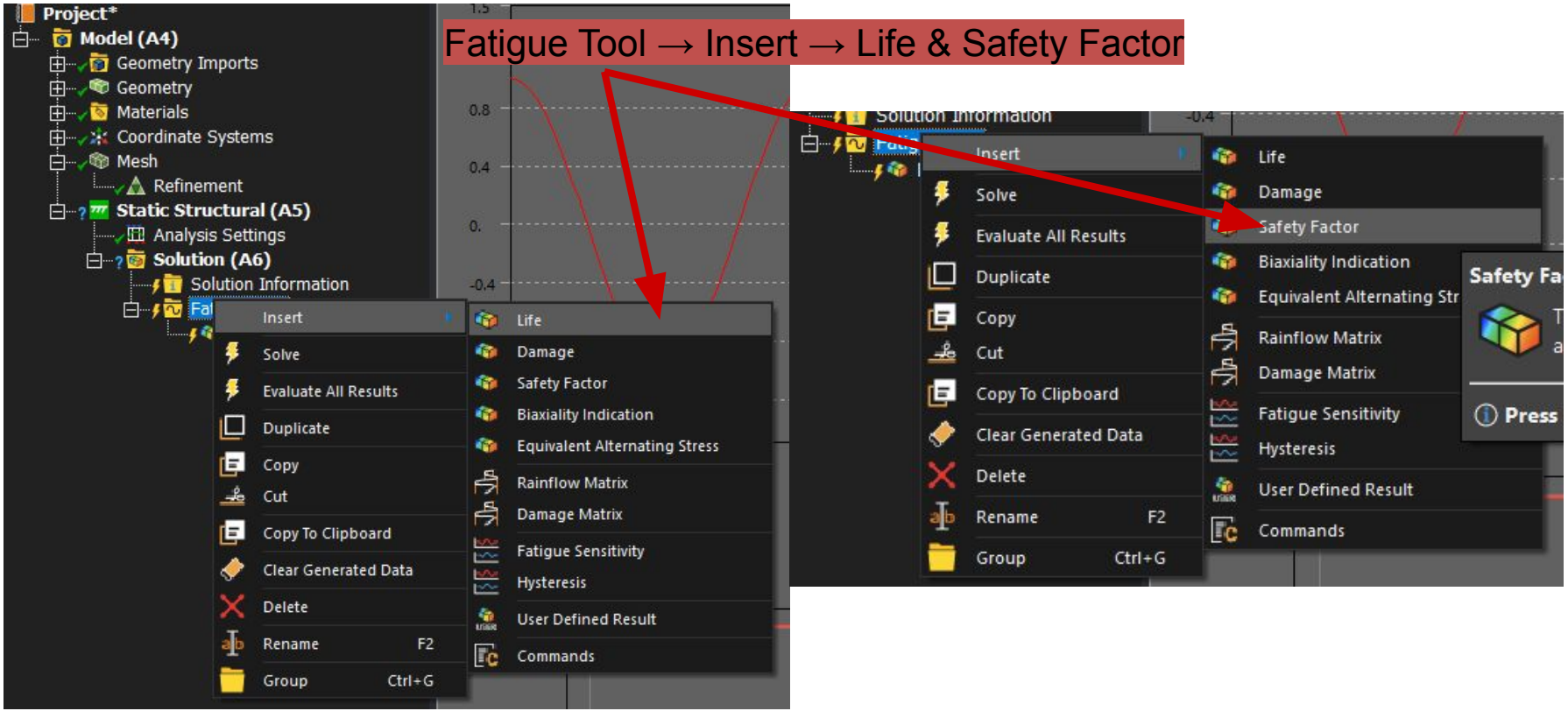


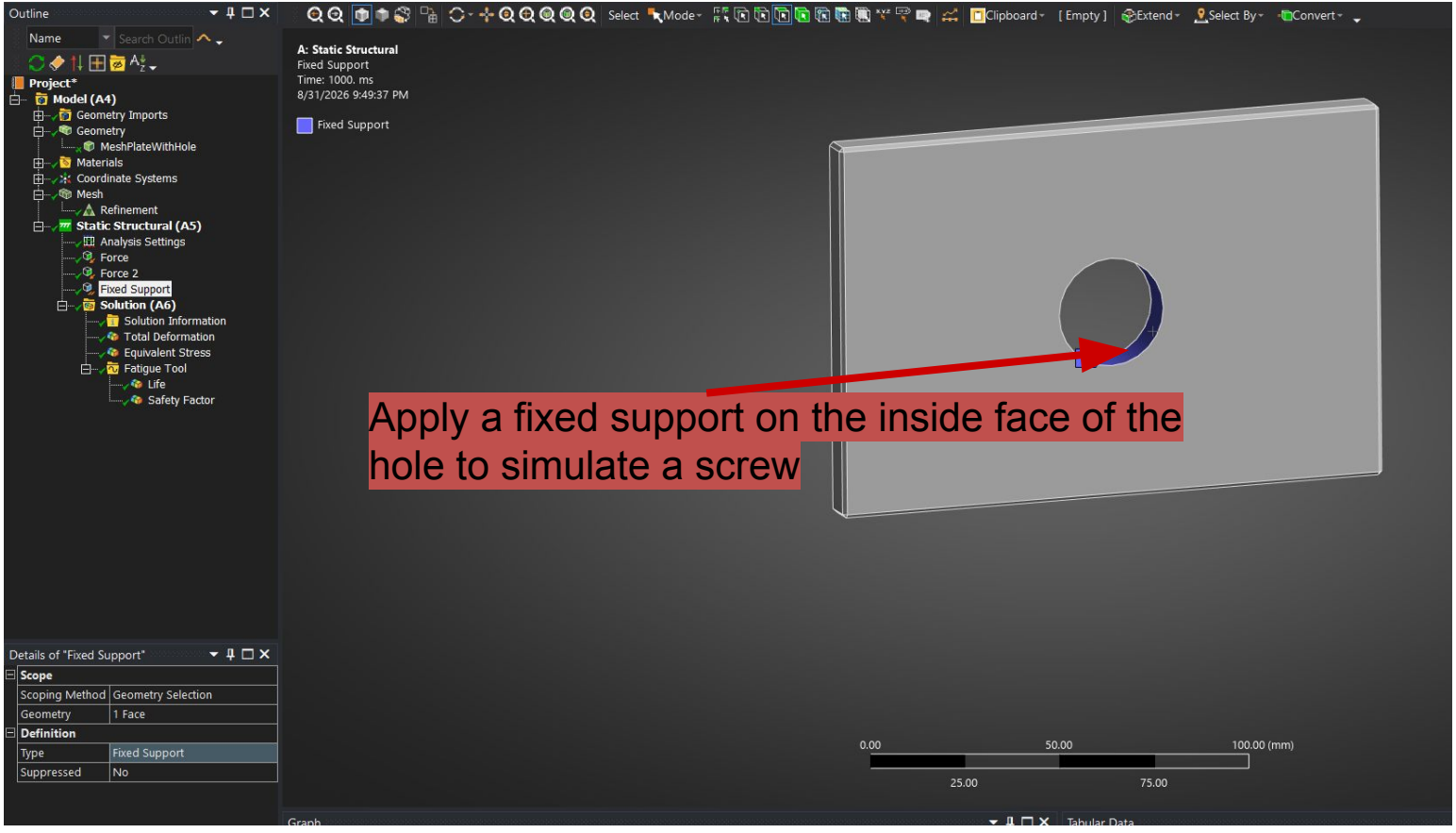
Hit “Fatigue Analysis” thru MA Wizard

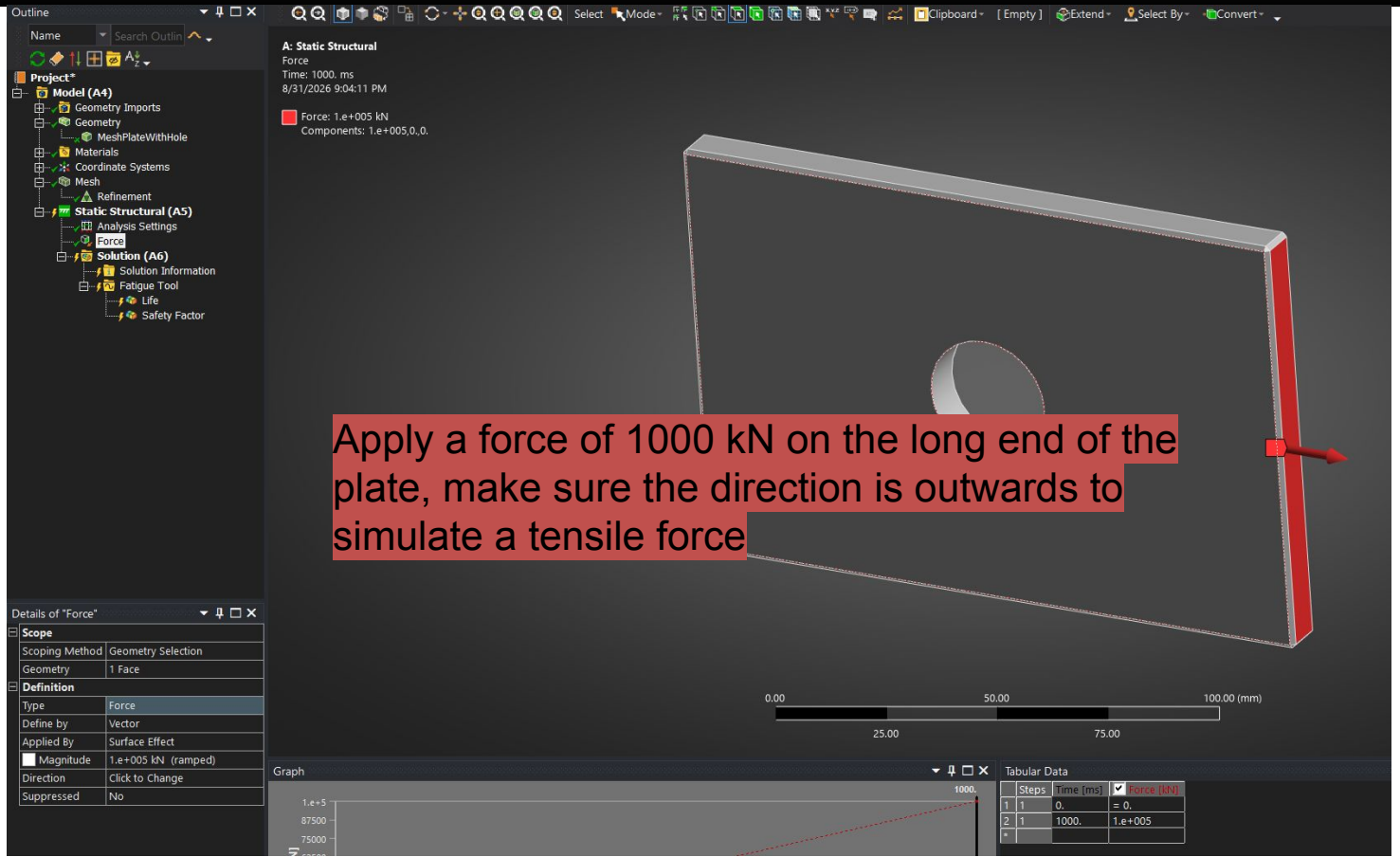
MA Wizard helps walk you through how to make a proper fatigue analysis on a part. Yellow means incomplete, green check is complete

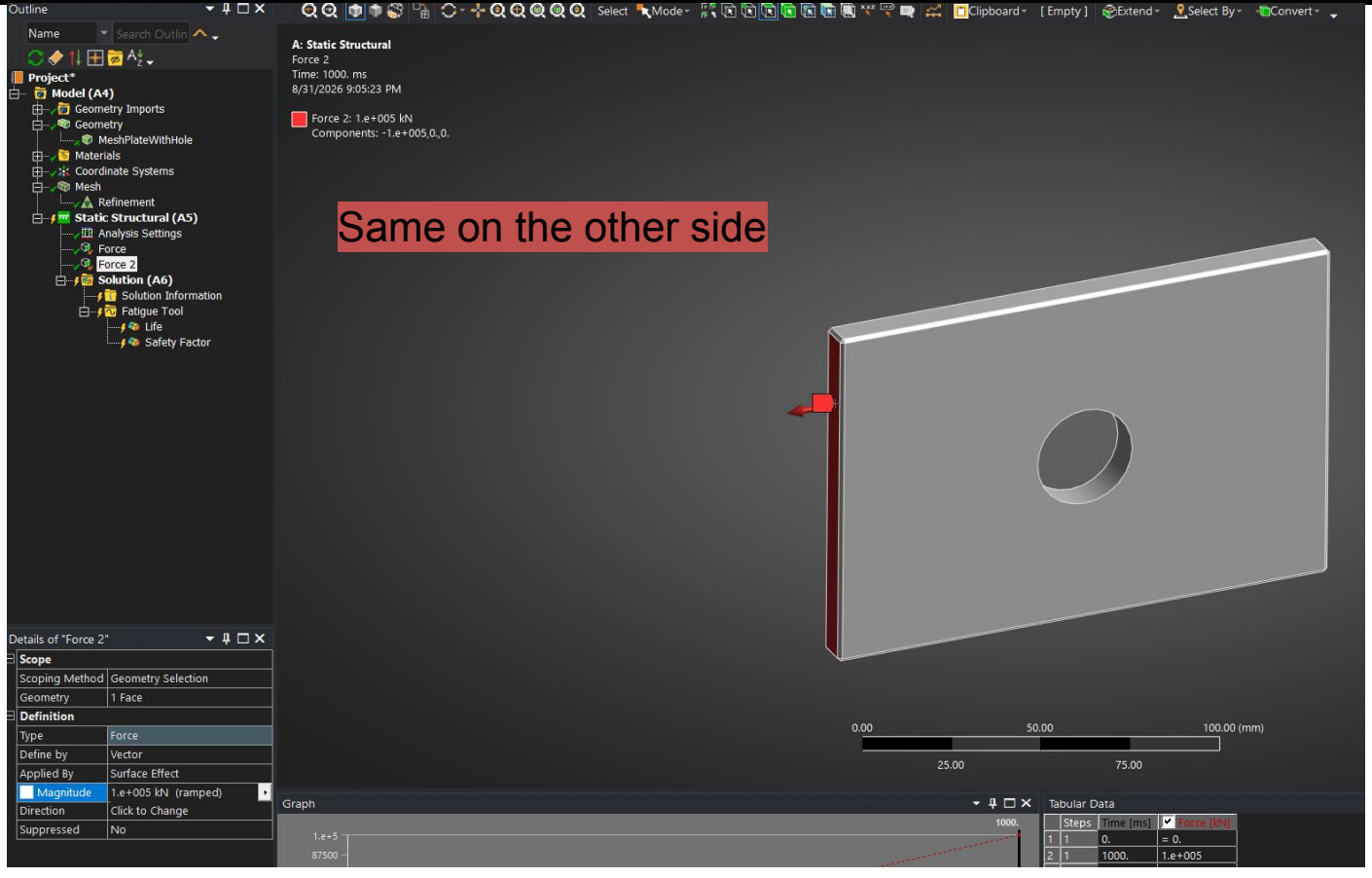

Project*
Model (A4)
Geometry Imports
Geometry
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Solution (
Solutic
Time: 1000. ms
8/31/2026 8:54:39 PM
Solution→Insert→Fatigue→Fatigue Tool
Insert
Solve
Evaluate All Results
Clear Generated Data
Rename F2
Group All Similar Children
Open Solver Files Directory
Deformation
Strain
Stress
Energy
Linearized Stress
Stress Tool
Fatigue
Contact Tool
Bolt Tool
Fracture Tool
Probe
Coordinate Systems
Volume
User Defined Result
Python Result
User Defined Criteria
Commands
Python Code
Fatigue Tool
Fatigue Tool
Fatigue provides life, damage, and safety factor information. Uses a stress-life or strain-life approach, with several options for handling mean stress and specifying loading conditions.
Press F1 for help.
Details of "Solution (A6)"

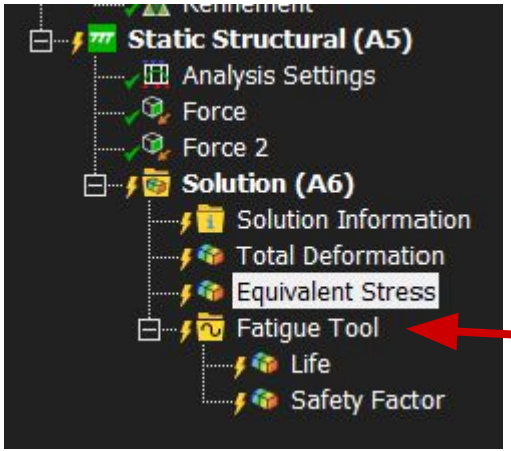

Outline
Name
Search Outline
Project*
Model (A4)
Geometry Imports
Geometry
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Solution (A6)
Solution Information
Fatigue Tool
Details of "Fatigue Tool"
Domain
Domain Type
Time
Materials
Fatigue Strength Factor (Kf)
1.
Loading
Type
Fully Reversed
Scale Factor
1.
Definition
Display Time
End Time
Options
Analysis Type
Stress Life
Mean Stress Theory
None
Stress Component
Equivalent (von-Mis...
Life Units
Units Name
cycles
1 cycle is equal to
1. cycles
Worksheet
Constant Amplitude Load
Fully Reversed
1.5
1.2
0.8
0.4
0.
-0.4
-0.8
-1.2
-1.5
Mean Stress Correction Theory
SN-None
Goodman
Soderberg
Gerber
ASME Elliptical
Endurance
0
Yield
Ultimate
Geometry
Worksheet
Graph
Tabular Data


Fatigue Tool → Insert → Life & Safety Factor
Project*
Model (A4)
Geometry Imports
Geometry
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Solution (A6)
Solution Information
Insert
Solve
Evaluate All Results
Duplicate
Copy
Cut
Copy To Clipboard
Clear Generated Data
Delete
Rename F2
Group Ctrl+G
Life
Damage
Safety Factor
Biaxiality Indication
Equivalent Alternating Stress
Rainflow Matrix
Damage Matrix
Fatigue Sensitivity
Hysteresis
User Defined Result
Commands
Safety Fa
Press


Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
MeshPlateWithHole
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Force
Force 2
Fixed Support
Solution (A6)
Solution Information
Total Deformation
Equivalent Stress
Fatigue Tool
Life
Safety Factor
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Fixed Support
Time: 1000. ms
8/31/2026 9:49:37 PM
Fixed Support
Apply a fixed support on the inside face of the hole to simulate a screw
0.00
50.00
100.00 (mm)
25.00
75.00
Details of "Fixed Support"
Scope
Scoping Method
Geometry Selection
Geometry
1 Face
Definition
Type
Fixed Support
Suppressed
No
Graph
Tabular Data


Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
MeshPlateWithHole
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Force
Solution (A6)
Solution Information
Fatigue Tool
Life
Safety Factor
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Force
Time: 1000. ms
8/31/2026 9:04:11 PM
Force: 1.e+005 kN
Components: 1.e+005,0.,0.
Apply a force of 1000 kN on the long end of the plate, make sure the direction is outwards to simulate a tensile force
0.00
50.00
100.00 (mm)
25.00
75.00
Details of "Force"
Scope
Scoping Method
Geometry Selection
Geometry
1 Face
Definition
Type
Force
Define by
Vector
Applied By
Surface Effect
Magnitude
1.e+005 kN (ramped)
Direction
Click to Change
Suppressed
No
Graph
1000.
1.e+5
87500
75000
Tabular Data
Steps
Time [ms]
Force [kN]
1
1
0.
= 0.
2
1
1000.
1.e+005

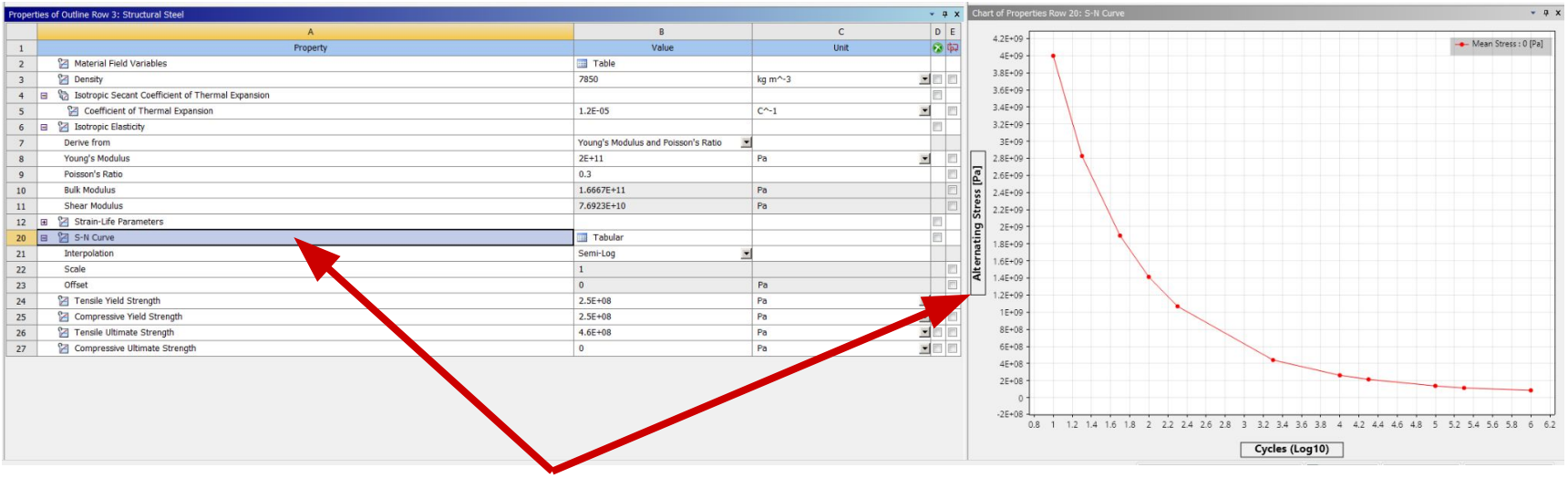

Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
MeshPlateWithHole
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Force
Force 2
Solution (A6)
Solution Information
Fatigue Tool
Life
Safety Factor
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Force 2
Time: 1000. ms
8/31/2026 9:05:23 PM
Force 2: 1.e+005 kN
Components: -1.e+005,0.,0.
Same on the other side
0.00
50.00
100.00 (mm)
25.00
75.00
Details of "Force 2"
Scope
Scoping Method
Geometry Selection
Geometry
1 Face
Definition
Type
Force
Define by
Vector
Applied By
Surface Effect
Magnitude
1.e+005 kN (ramped)
Direction
Click to Change
Suppressed
No
Graph
1000.
1.e+5
87500
Tabular Data
Steps
Time [ms]
Force [kN]
1
1
0.
= 0.
2
1
1000.
1.e+005



Details of "Fatigue Tool"

Domain	
Domain Type	Time
Materials	
Fatigue Strength Factor (Kf)	1.
Loading	
Type	Fully Reversed
Scale Factor	1.
Definition	
Display Time	End Time
Options	
Analysis Type	Stress Life
Mean Stress Theory	Goodman
Stress Component	Equivalent (von-Mis...
Life Units	
Units Name	cycles
1 cycle is equal to	1. cycles

Go to Fatigue Tool, set Mean Stress Theory to Goodman

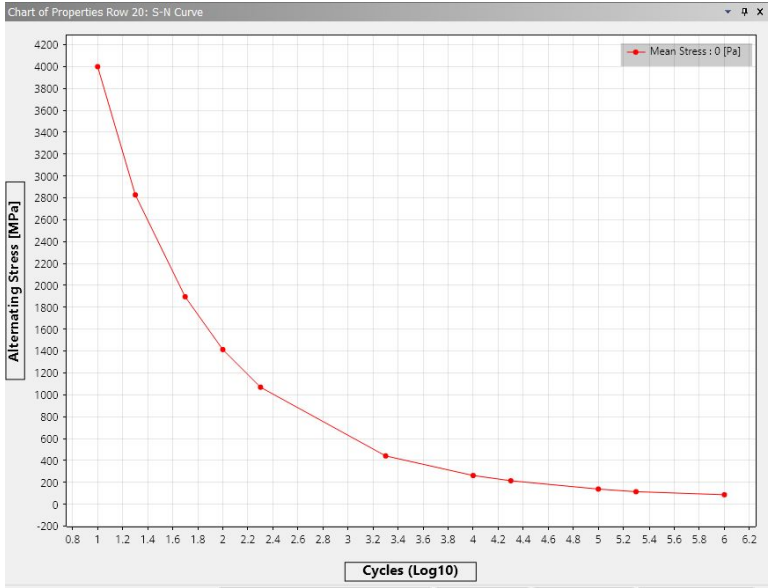


Properties of Outline Row 3: Structural Steel

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Density	7850	kg m^-3		
4	Isotropic Secant Coefficient of Thermal Expansion				
5	Coefficient of Thermal Expansion	1.2E-05	C^-1		
6	Isotropic Elasticity				
7	Derive from	Young's Modulus and Poisson's Ratio			
8	Young's Modulus	2E+11	Pa		
9	Poisson's Ratio	0.3			
10	Bulk Modulus	1.6667E+11	Pa		
11	Shear Modulus	7.6923E+10	Pa		
12	Strain-Life Parameters				
20	S-N Curve	Tabular			
21	Interpolation	Semi-Log			
22	Scale	1			
23	Offset	0	Pa		
24	Tensile Yield Strength	2.5E+08	Pa		
25	Compressive Yield Strength	2.5E+08	Pa		
26	Tensile Ultimate Strength	4.6E+08	Pa		
27	Compressive Ultimate Strength	0	Pa		

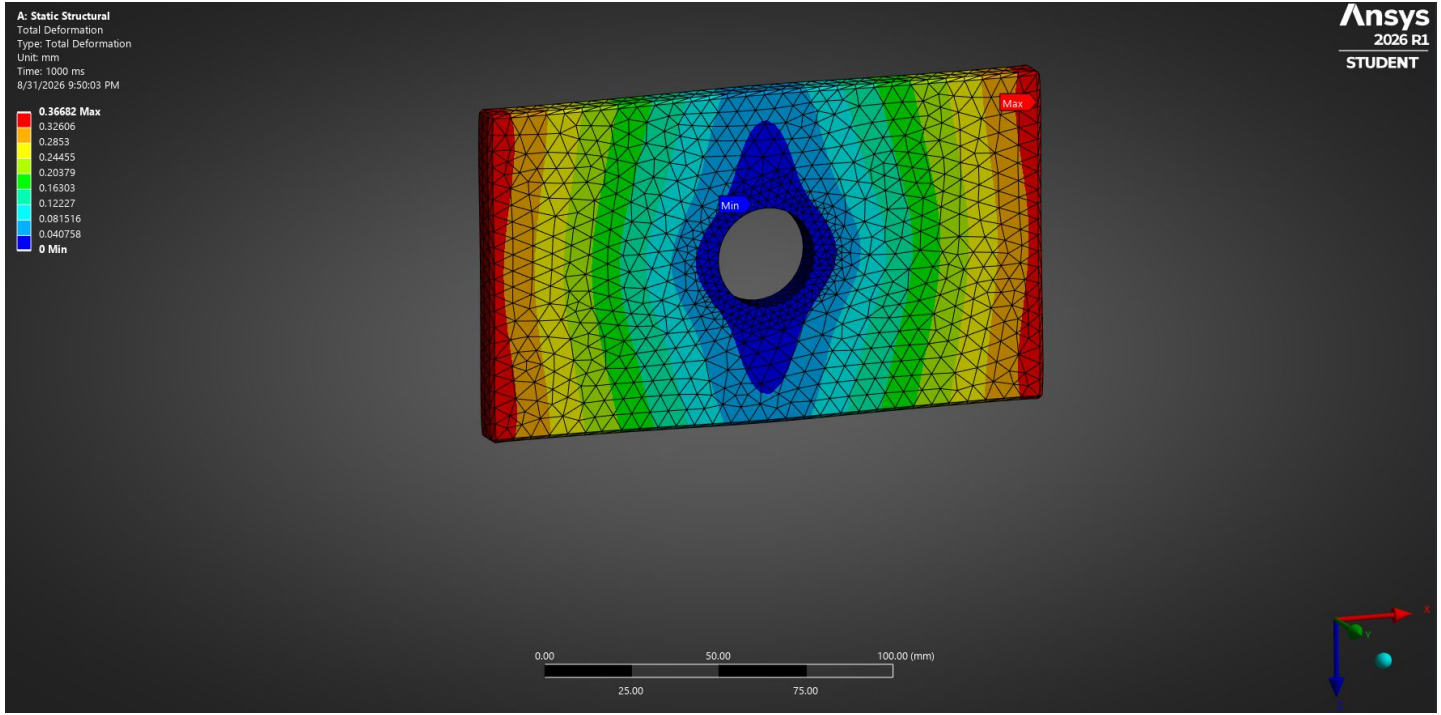
The “SN Curve” or stress-number curve plots the magnitude of alternating stress (S) against number of cycles to failure (N). This shows you how long a mechanical part will last under cyclic loading until breaking due to fatigue.

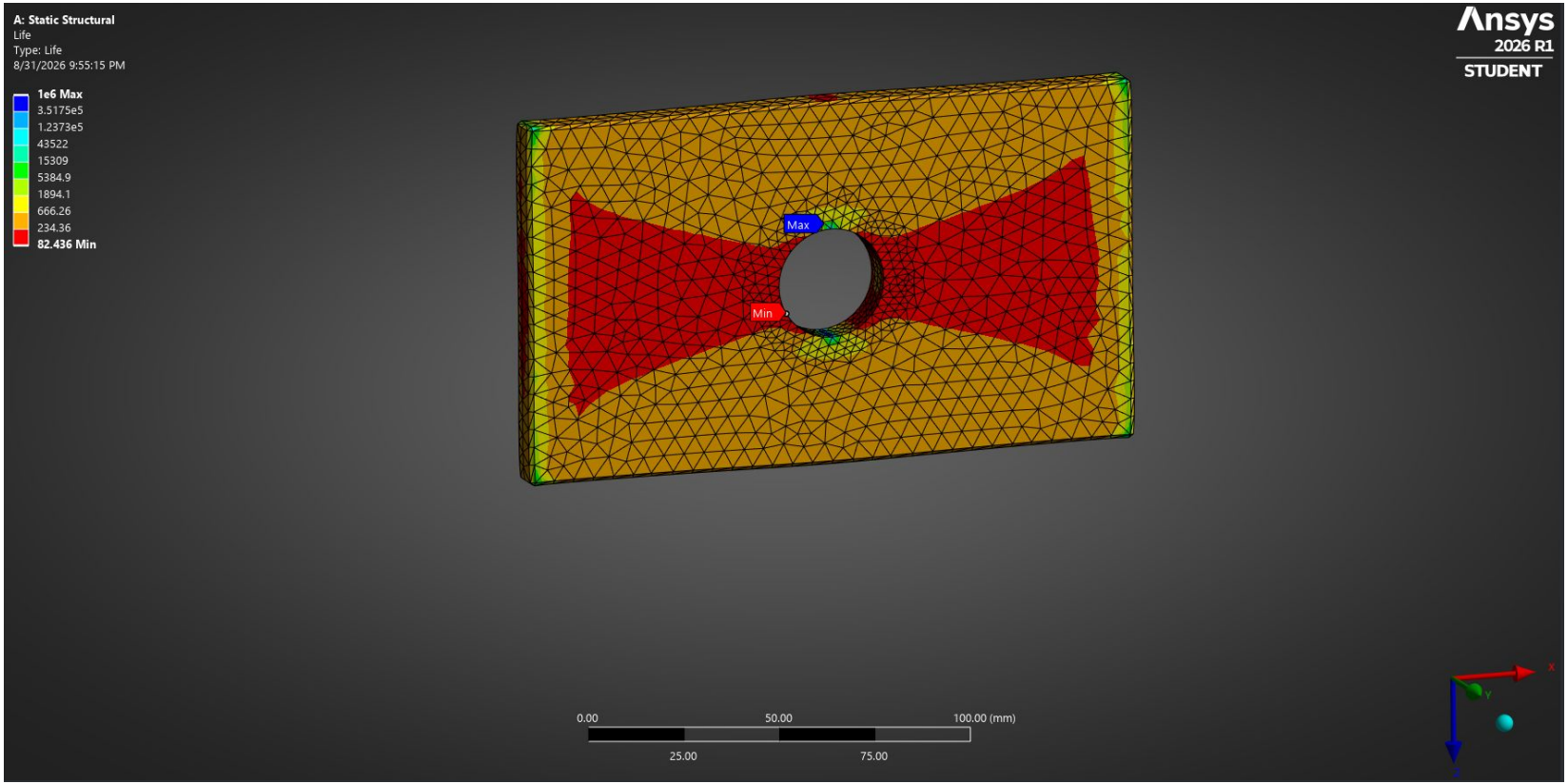
	B	C
1	Cycles	Alternating Stress (MPa)
2	10	3999
3	20	2827
4	50	1896
5	100	1413
6	200	1069
7	2000	441
8	10000	262
9	20000	214
10	1E+05	138
11	2E+05	114
12	1E+06	86.2
*		

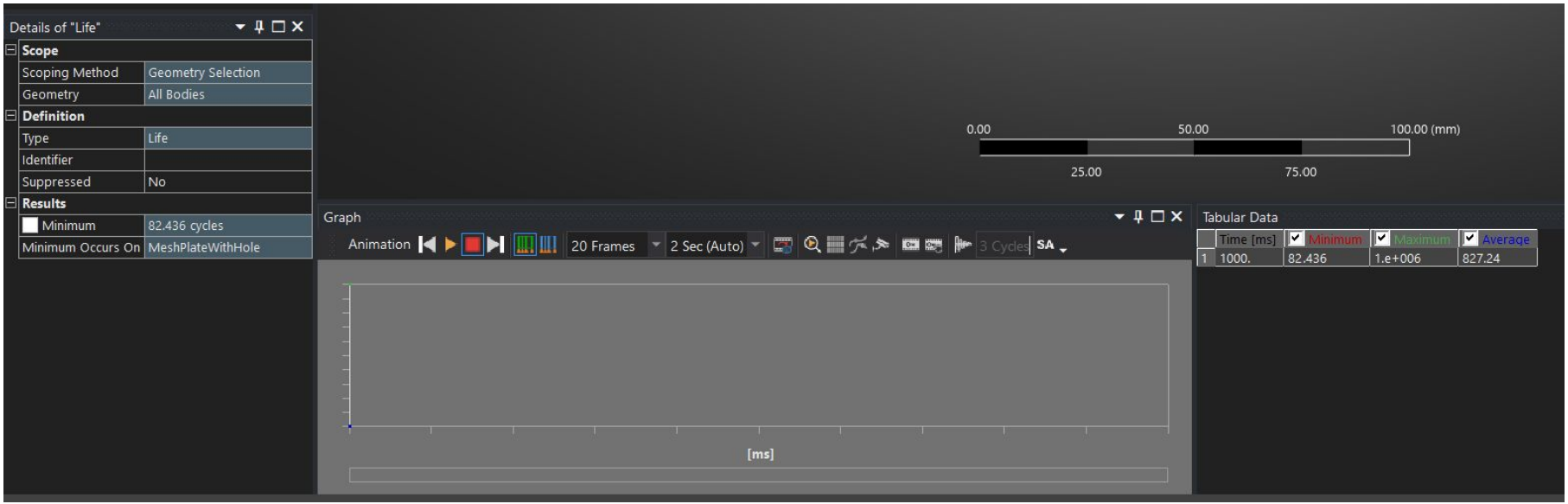


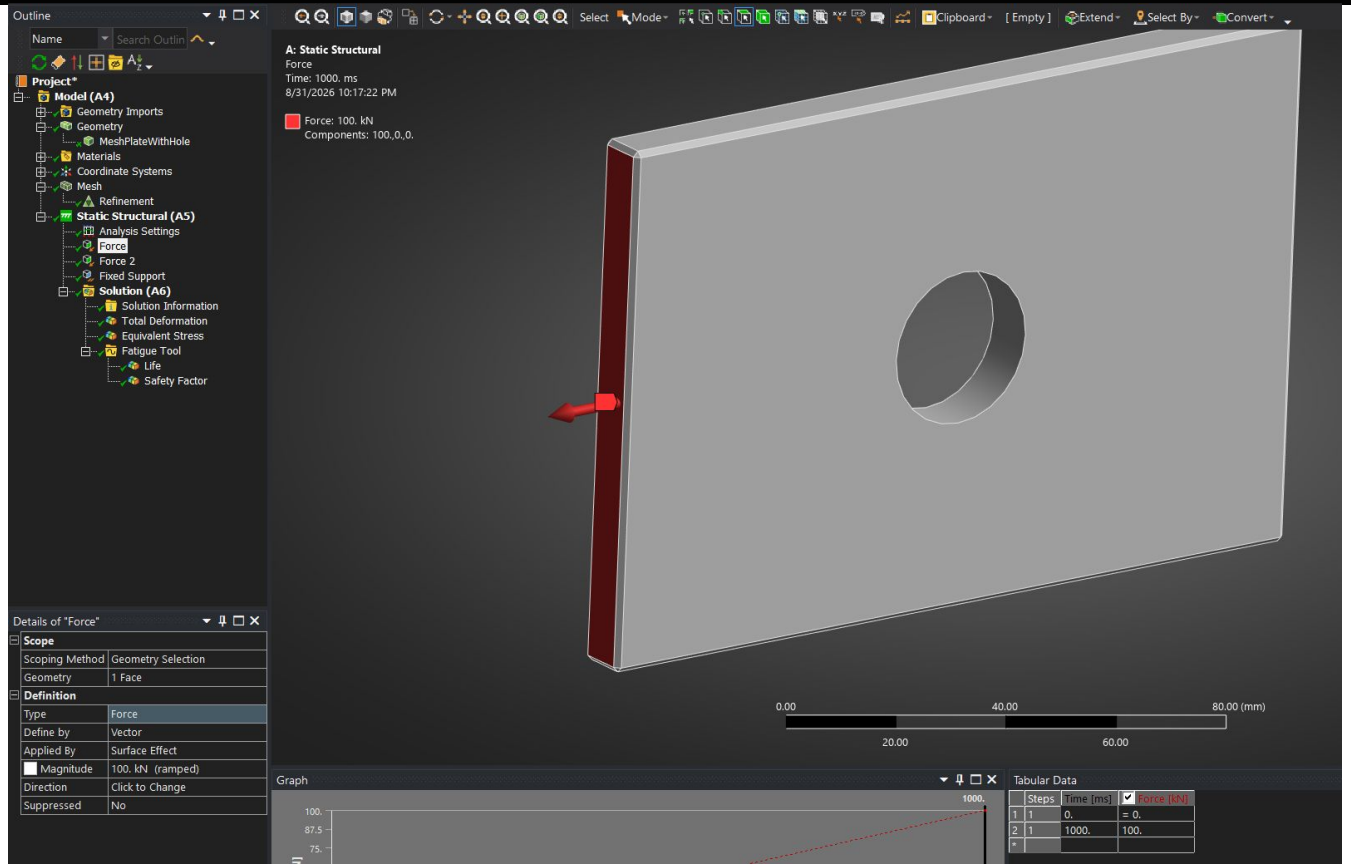
Endurance Limit, any more stress and it would not be safe. Loading conditions below this value is safe. As long as stress never reaches this value, material will never fail due to fatigue.

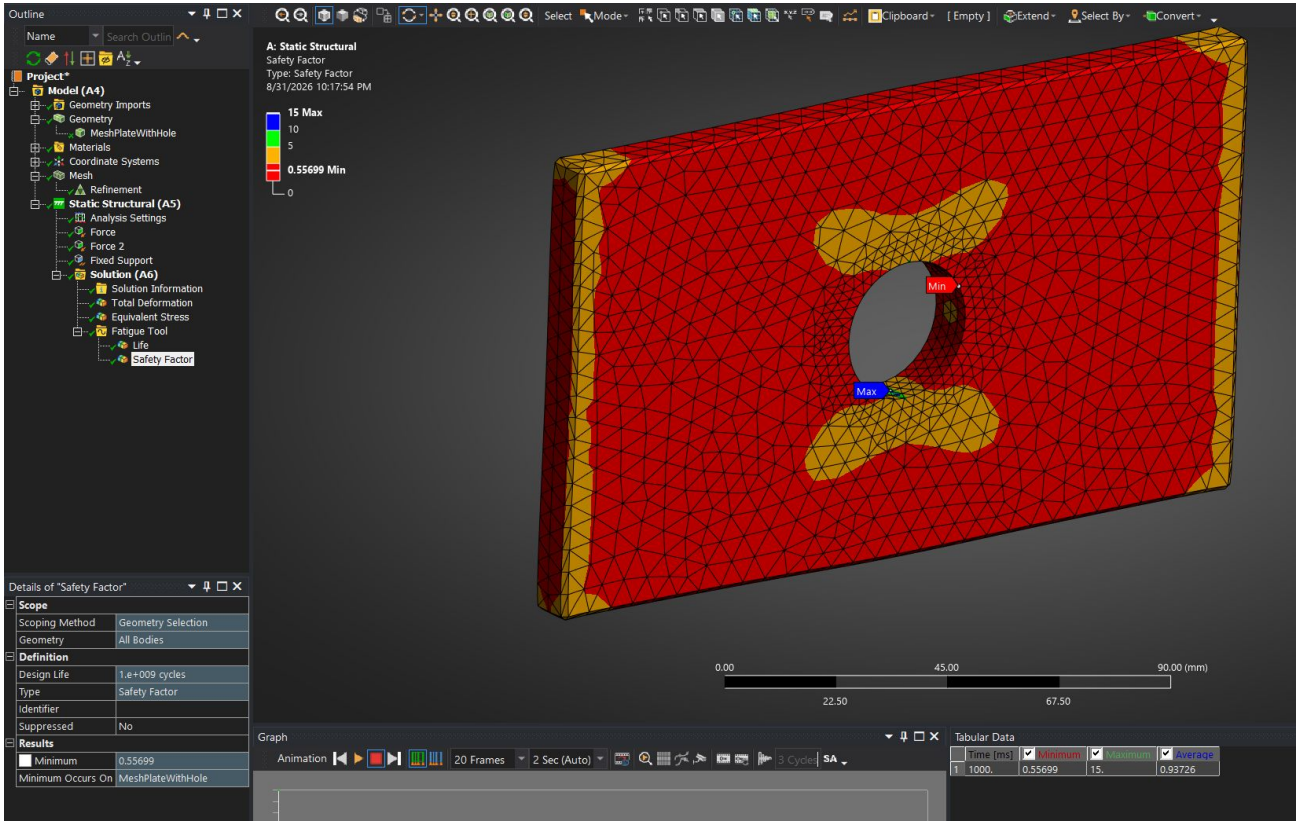

Solve
Solve

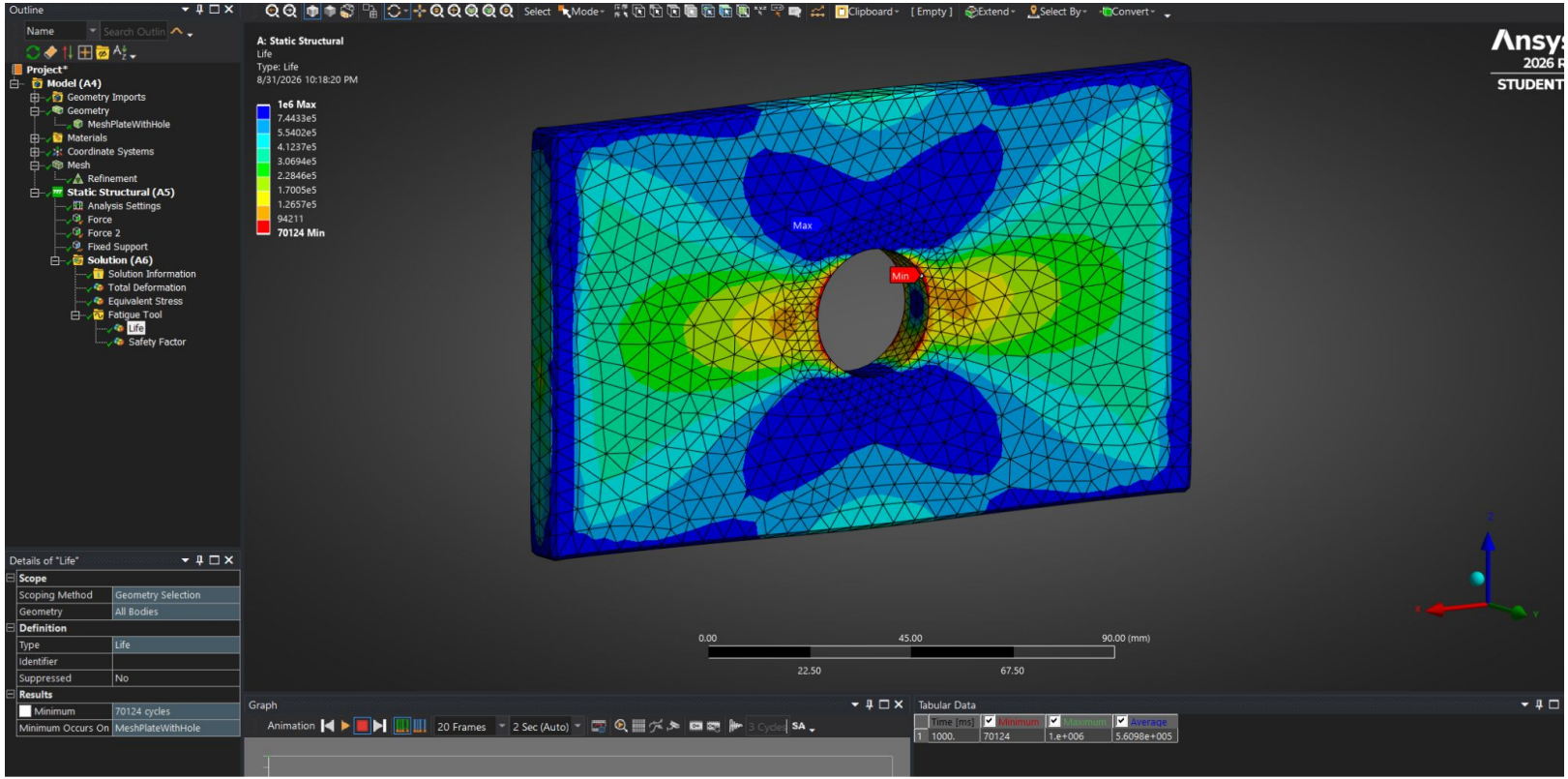

A: Static Structural
Total Deformation
Type: Total Deformation
Unit: mm
Time: 1000 ms
8/31/2026 9:50:03 PM
0.36682 Max
0.32606
0.2853
0.24455
0.20379
0.16303
0.12227
0.081516
0.040758
0 Min
Max
Min
Ansys
2026 R1
STUDENT
0.00
50.00
100.00 (mm)
25.00
75.00

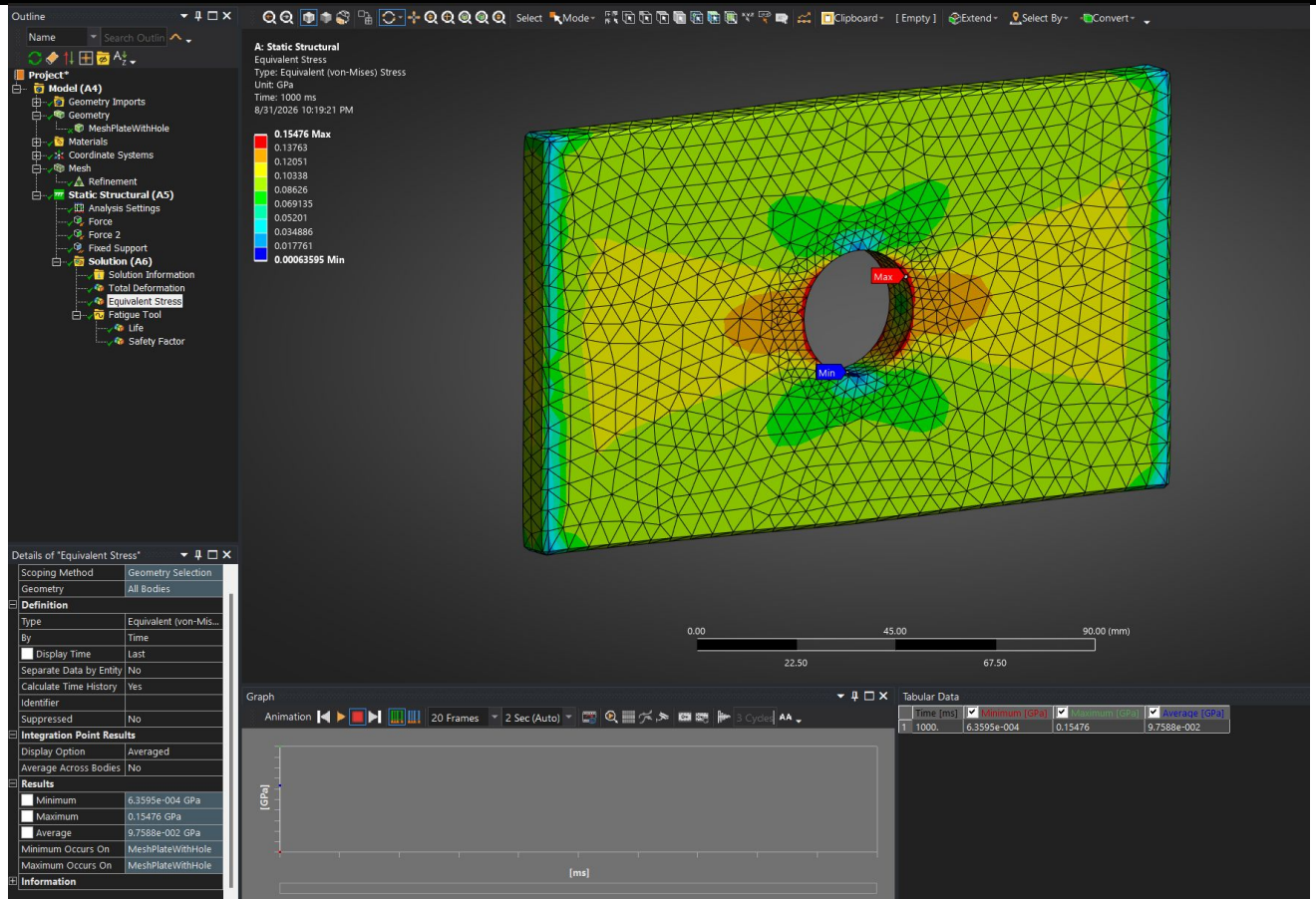

A: Static Structural
Life
Type: Life
8/31/2026 9:55:15 PM
1e6 Max
3.5175e5
1.2373e5
43522
15309
5384.9
1894.1
666.26
234.36
82.436 Min
Max
Min
Ansys
2026 R1
STUDENT
0.00
50.00
100.00 (mm)
25.00
75.00


Details of "Life"
Scope
Scoping Method
Geometry Selection
Geometry
All Bodies
Definition
Type
Life
Identifier
Suppressed
No
Results
Minimum
82.436 cycles
Minimum Occurs On
MeshPlateWithHole
0.00
50.00
100.00 (mm)
25.00
75.00
Graph
Animation
20 Frames
2 Sec (Auto)
3 Cycles
SA
[ms]
Tabular Data
Time [ms]
Minimum
Maximum
Average
1
1000.
82.436
1.e+006
827.24


Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
MeshPlateWithHole
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Force
Force 2
Fixed Support
Solution (A6)
Solution Information
Total Deformation
Equivalent Stress
Fatigue Tool
Life
Safety Factor
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Force
Time: 1000. ms
8/31/2026 10:17:22 PM
Force: 100. kN
Components: 100.,0.,0.
0.00
40.00
80.00 (mm)
20.00
60.00
Details of "Force"
Scope
Scoping Method
Geometry Selection
Geometry
1 Face
Definition
Type
Force
Define by
Vector
Applied By
Surface Effect
Magnitude
100. kN (ramped)
Direction
Click to Change
Suppressed
No
Graph
1000.
100.
87.5
75.
Tabular Data
Steps
Time [ms]
Force [kN]
1
1
0.
= 0.
2
1
1000.
100.


Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
MeshPlateWithHole
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Force
Force 2
Fixed Support
Solution (A6)
Solution Information
Total Deformation
Equivalent Stress
Fatigue Tool
Life
Safety Factor
Details of "Safety Factor"
Scope
Scoping Method
Geometry Selection
Geometry
All Bodies
Definition
Design Life
1.e+009 cycles
Type
Safety Factor
Identifier
Suppressed
No
Results
Minimum
0.55699
Minimum Occurs On
MeshPlateWithHole
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Safety Factor
Type: Safety Factor
8/31/2026 10:17:54 PM
15 Max
10
5
0.55699 Min
0
Min
Max
0.00
45.00
90.00 (mm)
22.50
67.50
Graph
Animation
20 Frames
2 Sec (Auto)
Tabular Data
Time [ms]
Minimum
Maximum
Average
1
1000.
0.55699
15.
0.93726


Outline
Name
Search Outlin
Project*
Model (A4)
Geometry Imports
Geometry
MeshPlateWithHole
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Force
Force 2
Fixed Support
Solution (A6)
Solution Information
Total Deformation
Equivalent Stress
Fatigue Tool
Life
Safety Factor
Details of "Life"
Scope
Scoping Method
Geometry Selection
Geometry
All Bodies
Definition
Type
Life
Identifier
Suppressed
No
Results
Minimum
70124 cycles
Minimum Occurs On
MeshPlateWithHole
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Life
Type: Life
8/31/2026 10:18:20 PM
1e6 Max
7.4433e5
5.5402e5
4.1237e5
3.0694e5
2.2846e5
1.7005e5
1.2657e5
94211
70124 Min
Max
Min
Ansys
2026 R
STUDENT
0.00
45.00
90.00 (mm)
22.50
67.50
Graph
Animation
20 Frames
2 Sec (Auto)
Tabular Data
Time [ms]
Minimum
Maximum
Average
1
1000.
70124
1.e+006
5.6098e+005


Outline
Name
Search Outline
Project*
Model (A4)
Geometry Imports
Geometry
MeshPlateWithHole
Materials
Coordinate Systems
Mesh
Refinement
Static Structural (A5)
Analysis Settings
Force
Force 2
Fixed Support
Solution (A6)
Solution Information
Total Deformation
Equivalent Stress
Fatigue Tool
Life
Safety Factor
Select
Mode
Clipboard
[Empty]
Extend
Select By
Convert
A: Static Structural
Equivalent Stress
Type: Equivalent (von-Mises) Stress
Unit: GPa
Time: 1000 ms
8/31/2026 10:19:21 PM
0.15476 Max
0.13763
0.12051
0.10338
0.08626
0.069135
0.05201
0.034886
0.017761
0.00063595 Min
Max
Min
0.00
45.00
90.00 (mm)
22.50
67.50
Details of "Equivalent Stress"
Scoping Method
Geometry Selection
Geometry
All Bodies
Definition
Type
Equivalent (von-Mis...
By
Time
Display Time
Last
Separate Data by Entity
No
Calculate Time History
Yes
Identifier
Suppressed
No
Integration Point Results
Display Option
Averaged
Average Across Bodies
No
Results
Minimum
6.3595e-004 GPa
Maximum
0.15476 GPa
Average
9.7588e-002 GPa
Minimum Occurs On
MeshPlateWithHole
Maximum Occurs On
MeshPlateWithHole
Information
Graph
Animation
20 Frames
2 Sec (Auto)
3 Cycles
[GPa]
[ms]
Tabular Data
Time [ms]
Minimum [GPa]
Maximum [GPa]
Average [GPa]
1
1000.
6.3595e-004
0.15476
9.7588e-002

TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Explicit Dynamic

https://www.youtube.com/watch?v=EhTO0_ovhnA

Why Explicit Dynamics?

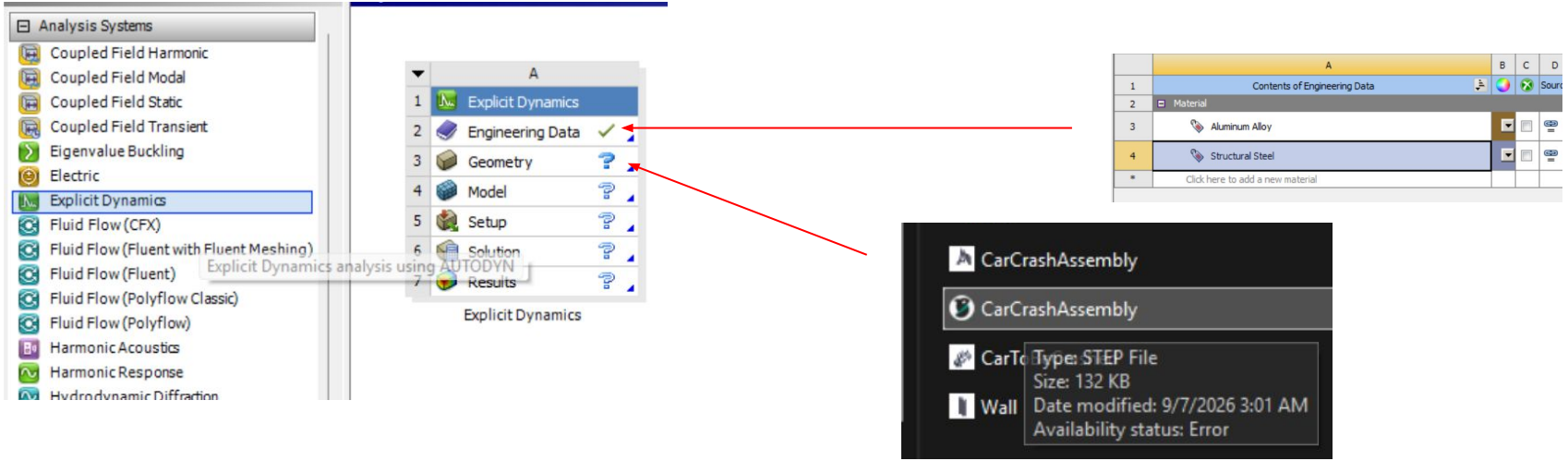


- Explicit dynamics can capture complex and nonlinear behaviour such as high impact/ high plasticity events like a car crash
- Explicit dynamics is usually much less computationally expensive and much more easy to set up for a problem of a short duration (~0.01 seconds) compared to using an implicit method like a transient structural simulation

Car Crash Analysis

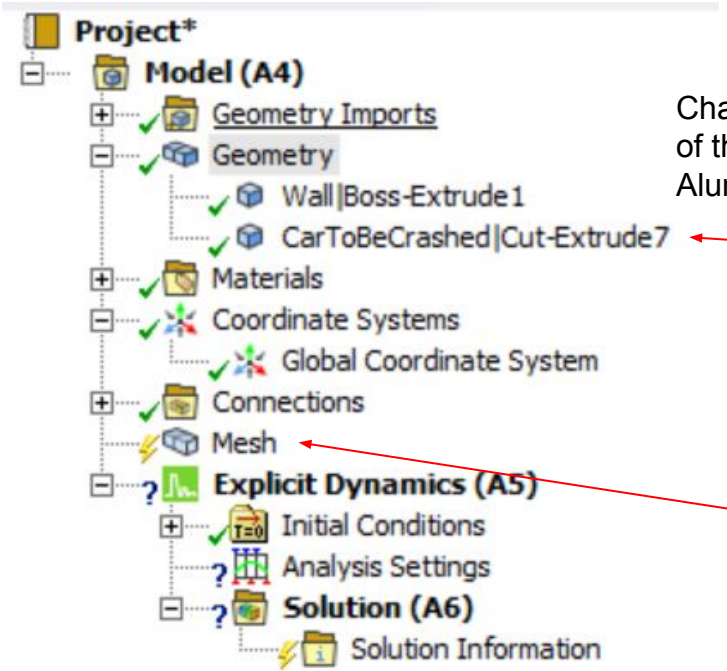
- Geometry: Car & wall
- Materials
  - Car: Aluminum Alloy
  - Wall: Structural Steel
- Boundary Conditions
  - Car: 50 m/s & Free Displacement in X & Z axis
  - Wall: Fixed
- Analysis Time = 0.01 seconds
  - Gives a travel time of 0.50 meters or 50 cm

Setting up geometry

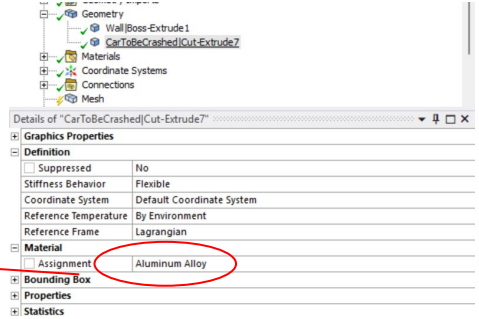


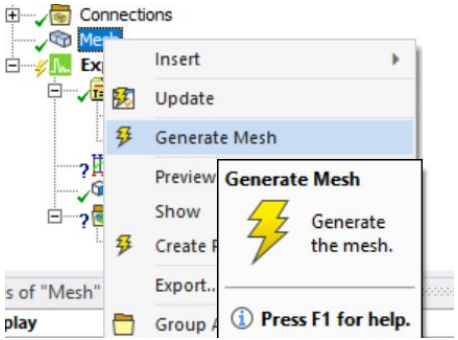
- Choose the explicit dynamics solver, import your car crash geometry, and add structural steel and aluminum alloy as your materials.

Set up



Change material of the car to Aluminum

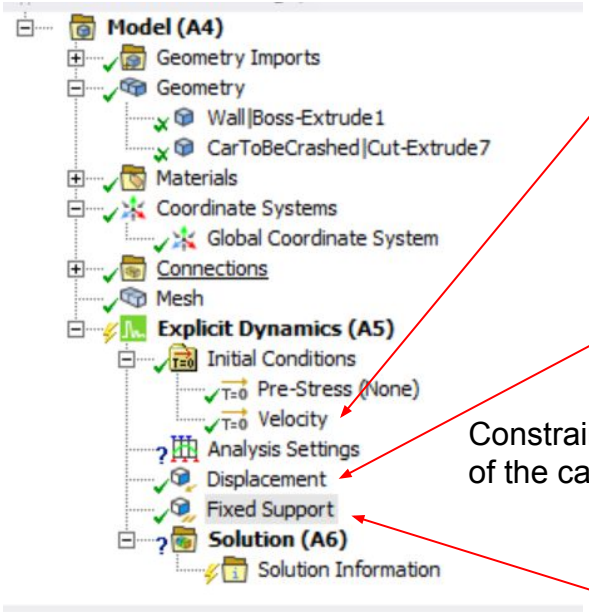




Generate the mesh

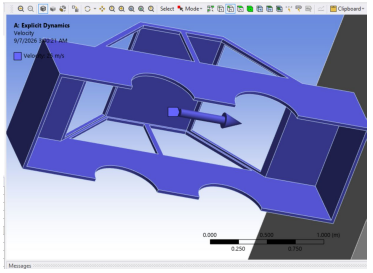
Conditions

Set the velocity of the vehicle to 50 m/s in the direction of the wall



Details of "Velocity"

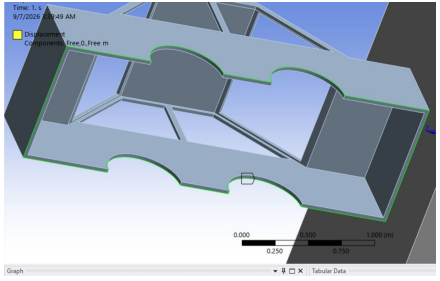
Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Input Type	Velocity
Pre-Stress Environment	None Available
Define by	Components
Coordinate System	Global Coordinate System
X Component	0. m/s
Y Component	0. m/s
Z Component	-50. m/s
Suppressed	No



Constrain the y-axis of the car

Details of "Displacement"

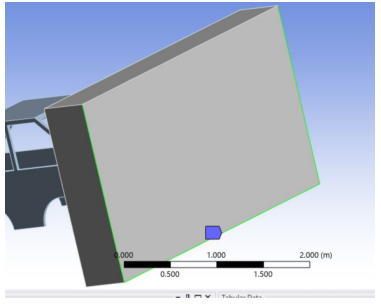
Scope		
Scoping Method	Geometry Selection	
Geometry	Apply	Cancel
Definition		
Type	Displacement	
Define by	Components	
Coordinate System	Global Coordinate System	
X Component	Free	
Y Component	0. m (step applied)	
Z Component	Free	
Suppressed	No	



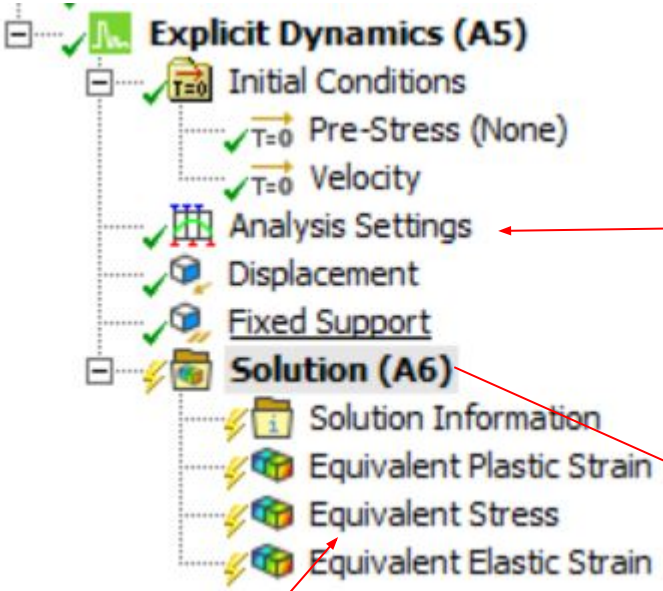
Details of "Fixed Support"

Scope		
Scoping Method	Geometry Selection	
Geometry	Apply	Cancel
Definition		
Type	Fixed Support	
Suppressed	No	

Fix the edges of the wall in place

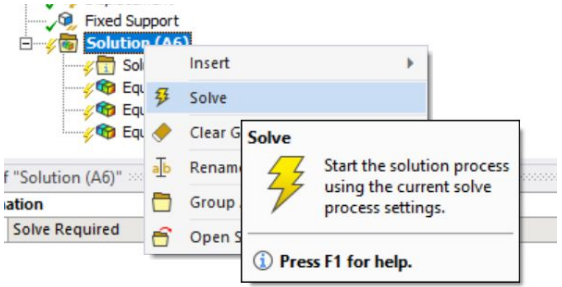


Solving



Set the end time of the simulation

Details of "Analysis Settings"	
Analysis Settings Preference	
Type	Program Controlled
Step Controls	
Number Of Steps	1
Current Step Number	1
Load Step Type	Explicit Time Integration
End Time	1.e-002 s
Resume From Cycle	0
Maximum Number of Cycles	1e+07
Maximum Energy Error	0.1
Reference Energy Cycle	0
Initial Time Step	Program Controlled
Minimum Time Step	Program Controlled
Maximum Time Step	Program Controlled
Time Step Safety Factor	0.9
Characteristic Dimension	Diagonals
Automatic Mass Scaling	No
Solver Controls	
Solve Units	mm, mg, ms

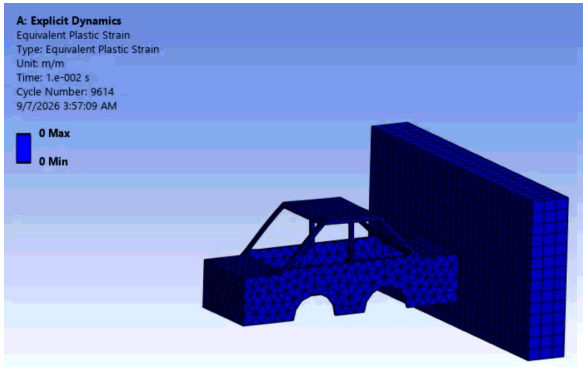


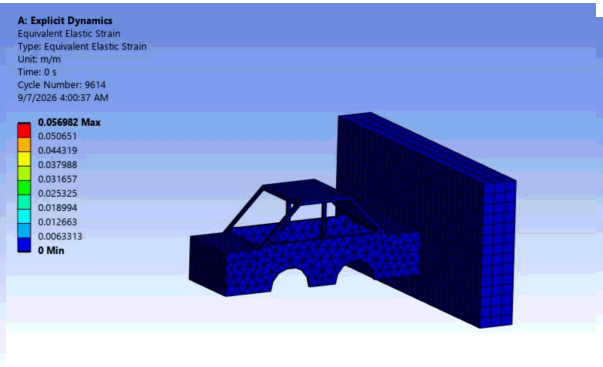
solve!

Add your intended solutions

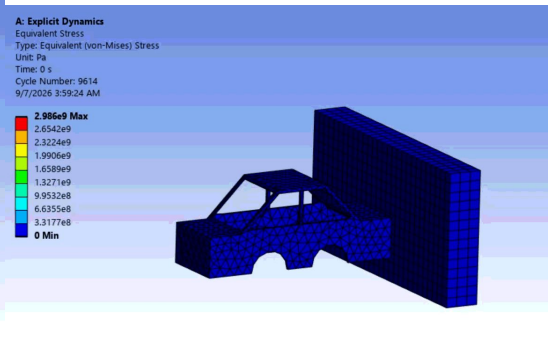
Solutions

In a 50 m/s car accident, this car has lots of elastic deformation, however there is very little plastic deformation





Stress is concentrated around the body of the car and the windows





“Who is my neighbor”

-John August Swanson

Incubator